EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

Department of Computer Science and Engineering

'Jnana Prabha', Virgo Nagar Post, Bengaluru-560049

Academic Year: 2024-25

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LABORATORY MANUAL

SEMESTER : VI

SUBJECT : Devops

SUBCODE : BCSL657D

NAME: _____

USN: _____

SECTION: _____

[Type here]

PROGRAM OUTCOMES

Engineering Graduates will able to:

Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

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Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and finance: Demonstrate knowledge and understanding of the Engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life -long learning: Recognize the need for and have the preparation and ability to engage in independent and life -long learning in the broadest context of technological change.

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EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

INSTITUTE VISION AND MISSION

VISION

The East Point College of Engineering and Technology aspires to be a globally acclaimed institution, recognized for excellence in engineering education, applied research and nurturing students for holistic development.

MISSION

M1: To create engineering graduates through quality education and to nurture innovation, creativity and excellence in teaching, learning and research

M2: To serve the technical, scientific, economic and societal developmental needs of our communities

M3: To induce integrity, teamwork, critical thinking, personality development and ethics in students and to lay the foundation for lifelong learning

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EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

Department of Computer Science and Engineering

DEPARTMENT VISION AND MISSION

VISION

The department aspires to be a Centre of excellence in Computer Science & Engineering to develop competent professionals through holistic development.

MISSION

M1: To create successful Computer Science Engineering graduates through effective pedagogies, the latest tools and technologies, and excellence in teaching and learning.

M2: To augment experiential learning skills to serve technical, scientific, economic, and social developmental needs.

M3: To instill integrity, critical thinking, personality development, and ethics in students for a successful career in Industries, Research, and Entrepreneurship.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1: To produce graduates who can perform technical roles to contribute effectively in software industries and R&D Centre

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PEO 2: To produce graduates having the ability to adapt and contribute in key domains of computer science and engineering to develop competent solutions.

PEO 3: To produce graduates who can provide socially and ethically responsible solutions while adapting to new trends in the domain to carve a successful career in the industry

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: To conceptualize, model, design, simulate, analyze, develop, test, and validate computing systems and solve technical problems arising in the field of computer science & engineering.

PSO2: To specialize in the sub-areas of computer science & engineering systems such as cloud computing, Robotic Process Automation, cyber security, big data analytics, user interface design, and IOT to meet industry requirements.

PSO3: To build innovative solutions to meet the demands of the industry using appropriate tools and techniques.

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COURSE LEARNING OBJECTIVES

Course objectives:

1. To introduce DevOps terminology, definition & concepts
2. To understand the different Version control tools like Git, Mercurial
3. To understand the concepts of Continuous Integration/ Continuous Testing/ Continuous Deployment)
4. To understand Configuration management using Ansible

5. Illustrate the benefits and drive the adoption of cloud-based Devops tools to solve real world problems

COURSE OUTCOMES

At the end of the course the student will be able to:

Course outcomes (Course Skill Set): At the end of the course, the student will be able to:

1. Demonstrate different actions performed through Version control tools like Git.
2. Perform Continuous Integration and Continuous Testing and Continuous Deployment using Jenkins by building and automating test cases using Maven & Gradle
3. Experiment with configuration management using Ansible.
4. Demonstrate Cloud-based DevOps tools using Azure DevOps.

Sl. NO	Experiments
1	Introduction to Maven and Gradle: Overview of Build Automation Tools, Key Differences Between Maven and Gradle, Installation and Setup
2	Working with Maven: Creating a Maven Project, Understanding the POM File, Dependency Management and Plugins
3	Working with Gradle: Setting Up a Gradle Project, Understanding Build Scripts (Groovy and Kotlin DSL), Dependency Management and Task Automation
4	Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle
5	Introduction to Jenkins: What is Jenkins?, Installing Jenkins on Local or Cloud Environment, Configuring Jenkins for First Use
6	Continuous Integration with Jenkins: Setting Up a CI Pipeline, Integrating Jenkins with Maven/Gradle, Running Automated Builds and Tests

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7	Configuration Management with Ansible: Basics of Ansible: Inventory, Playbooks, and Modules, Automating Server Configurations with Playbooks, Hands-On: Writing and Running a Basic Playbook
8	Practical Exercise: Set Up **a Jenkins CI Pipeline for a Maven Project**, Use Ansible to Deploy Artifacts Generated by Jenkins
9	Introduction to Azure DevOps: Overview of Azure DevOps Services, Setting Up an Azure DevOps Account and Project
10	Creating Build Pipelines: Building a Maven/Gradle Project with Azure Pipelines, Integrating Code Repositories (e.g., GitHub, Azure Repos), Running Unit Tests and Generating Reports
11	Creating Release Pipelines: Deploying Applications to Azure App Services, Managing Secrets and Configuration with Azure Key Vault, Hands-On: Continuous Deployment with Azure Pipelines
12	Creating Release Pipelines: Deploying Applications to Azure App Services, Managing Secrets and Configuration with Azure Key Vault, Hands-On: Continuous Deployment with Azure Pipelines

Assessment Details (both CIE and SEE) The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together Continuous Internal Evaluation (CIE): CIE marks for the practical course are 50 Marks. The split-up of CIE marks for record/ journal and test are in the ratio 60:40.

● Each experiment is to be evaluated for conduction with an observation sheet and record write-up. Rubrics for the evaluation of the journal/write-up for hardware/software experiments are designed by the faculty who is handling the laboratory session and are made known to students at the beginning of the practical session.

● Record should contain all the specified experiments in the syllabus and each experiment write-up will be evaluated for 10 marks.

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● Total marks scored by the students are scaled down to 30 marks (60% of maximum marks).

● Weightage to be given for neatness and submission of record/write-up on time.

● Department shall conduct a test of 100 marks after the completion of all the experiments listed in the syllabus.

● In a test, test write-up, conduction of experiment, acceptable result, and procedural knowledge will carry a weightage of 60% and the rest 40% for viva-voce.

● The suitable rubrics can be designed to evaluate each student's performance and learning ability.

● The marks scored shall be scaled down to 20 marks (40% of the maximum marks). The Sum of scaled-down marks scored in the report write-up/journal and marks of a test is the total CIE marks scored by the student

Semester End Evaluation (SEE):

● SEE marks for the practical course are 50 Marks.

● SEE shall be conducted jointly by the two examiners of the same institute, examiners are appointed by the Head of the Institute.

● The examination schedule and names of examiners are informed to the university before the conduction of the examination. These practical examinations are to be conducted between the schedule mentioned in the academic calendar of the University.

● All laboratory experiments are to be included for practical examination.

● (Rubrics) Breakup of marks and the instructions printed on the cover page of the answer script to be strictly adhered to by the examiners. OR based on the course requirement evaluation rubrics shall be decided jointly by examiners.

Students can pick one question (experiment) from the questions lot prepared by the examiners jointly.

● Evaluation of test write-up/ conduction procedure and result/viva will be conducted jointly by examiners.

● General rubrics suggested for SEE are mentioned here, writeup-20%, Conduction procedure and result in -60%, Viva-voce 20% of maximum marks. SEE for practical shall be evaluated for 100 marks and scored marks shall be scaled down to 50 marks (however, based on course type, rubrics shall be decided by the examiners) Change of experiment is allowed only

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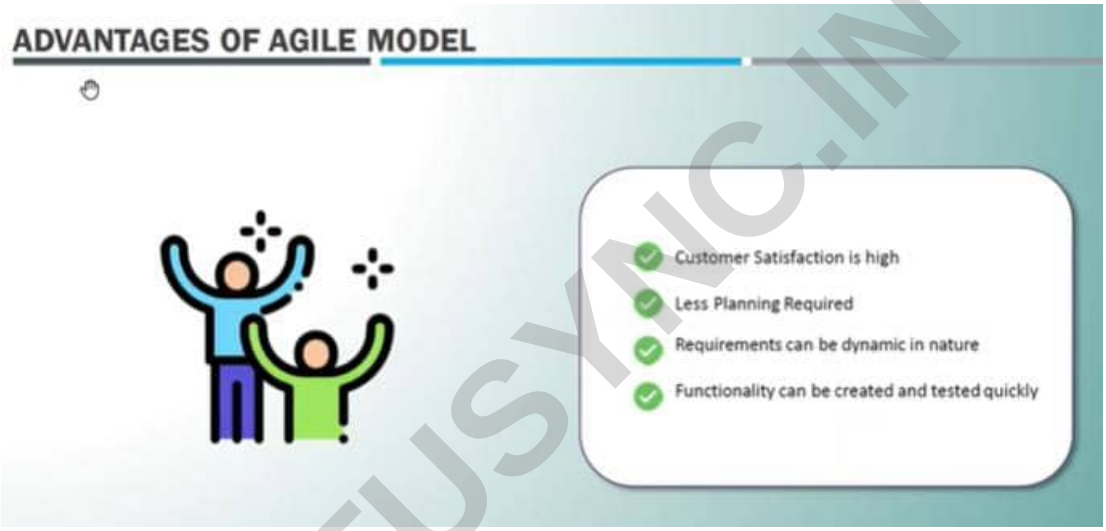
once and 15% of Marks allotted to the procedure part are to be made zero. The minimum duration of SEE is 02 hours

Suggested Learning Resources:

- https://www.geeksforgeeks.org/devops-tutorial/
- https://www.javatpoint.com/devops
- https://www.youtube.com/watch?v=2N-59wUIPVI
- https://www.youtube.com/watch?v=87ZqwoFeO88



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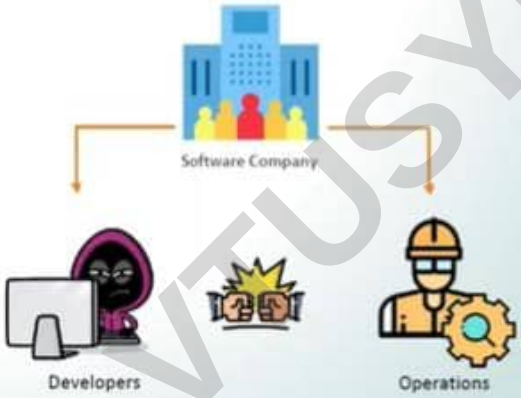
DISADVANTAGES OF AGILE MODEL

- Not suitable for handling complex dependencies inprojects
- Knowledge transfer to colleagues can be difficult since there is little documentation
- Success of the project depends heavily on customer interaction

WHY DEVOPS?



Although, the software quality was improved. We still had a lack of efficiency among the development team. A typical software development team consists of Developers and Operations employees. Let us understand their job roles

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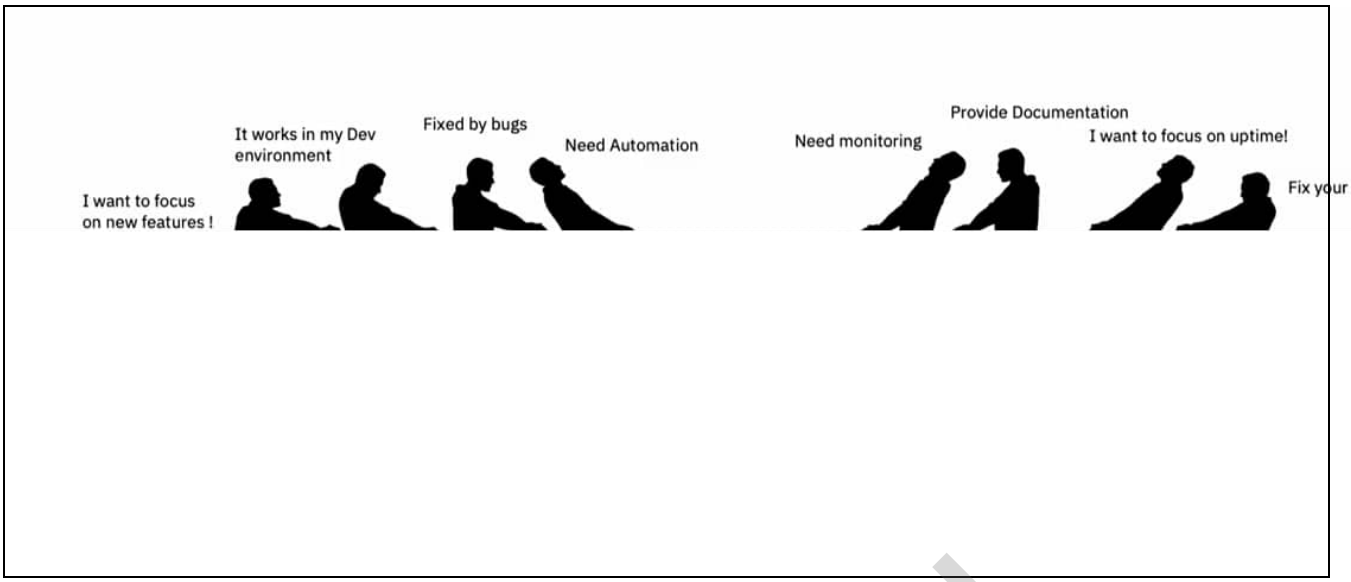
I want to focus
on new features !
It works in my Dev
environment
Fixed by bugs
Need Automation
Need monitoring
Provide Documentation
I want to focus on uptime!
Fix your bugs!

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1. Market Share & Popularity

1. **AWS**: The market leader, with around **32-34%** of the global cloud market.
2. **Microsoft Azure**: Second largest, with around **22-24%** market share.
3. **Google Cloud**: Third, with about **11-12%** market share.

AWS dominates in market adoption, while Azure benefits from its enterprise-friendly Microsoft integrations. Google Cloud is growing but remains behind AWS and Azure.

2. Services & Features

Feature	AWS	Microsoft Azure	Google Cloud (GCP)
Compute	EC2, Lambda	Virtual Machines, Functions	Compute Engine, Cloud Run
Storage	S3, EBS, Glacier	Blob Storage, Disk Storage	Cloud Storage, Persistent Disk
Database	RDS, DynamoDB, Redshift	SQL Database, Cosmos DB	Cloud SQL, Bigtable, Firestore
AI/ML	SageMaker	Azure AI, Machine Learning	TensorFlow, Vertex AI
Networking	VPC, CloudFront	Virtual Network, ExpressRoute	VPC, Cloud CDN
Security	IAM, GuardDuty, Shield	Active Directory, Sentinel	IAM, Security Command Center
Kubernetes	EKS	AKS	GKE (Most advanced for K8s)
Serverless	Lambda	Azure Functions	Cloud Functions

1. **AWS**: The most mature and feature-rich.
2. **Azure**: Best for hybrid cloud and Microsoft ecosystem users.
3. **GCP**: Excels in AI, big data, and Kubernetes.

3. Pricing

Pricing depends on usage, but here's a general comparison:

1. **AWS**: Pay-as-you-go, per-second billing. Can be complex but offers a free tier.
2. **Azure**: Similar to AWS, but often provides cost advantages for Windows-based businesses.
3. **Google Cloud**: Generally more cost-effective for compute and AI workloads.

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4. Ease of Use

1. **AWS**: Can be complex due to the vast number of services.
2. **Azure**: Easier for businesses already using Microsoft products.
3. **Google Cloud**: Simplified UI, developer-friendly.

5. Strengths & Weaknesses

Provider	Strengths	Weaknesses
AWS	Largest selection of services, global reach	Pricing complexity, can be costly
Azure	Best for enterprise & Microsoft integration	Can be difficult for non-Microsoft users
Google Cloud	Best for AI/ML, Kubernetes, analytics	Fewer global data centers, smaller service range

Conclusion

1. Choose **AWS** for the most comprehensive cloud ecosystem.
2. Choose **Azure** if you rely on Microsoft services.
3. Choose **Google Cloud** if your focus is AI, data, and Kubernetes.

Software Installations:

Method1 : Run through Command promt

Method : Run through IDE / VS Code or Inelli IDE

https://www.youtube.com/watch?v=Y6Qd_Bovo-o

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Program:1

How to Create and Run a Gradle Java Project in VS Code

1. Install Prerequisites

Before starting, make sure you have the following installed:

✅ **Java Development Kit (JDK 21+)**
✅ **Gradle** – Download & Install
✅ **Visual Studio Code**
✅ **VS Code Extensions**:

1. Install **Java Extension Pack**
2. Install **Gradle for Java**

2. Create a Gradle Java Project

Using Gradle CLI (Recommended)

1. **Open Terminal** and run:

sh

Copy Edit

gradle init

2. Select:
    1. **Project Type**: Application
    2. **Implementation Language**: Java
    3. **Build Script**: Groovy or Kotlin
    4. **Test Framework**: JUnit (or any other preference)
    5. **Project Name**: (default or custom)

This will generate a **Gradle Java project structure**.

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3. Open the Project in VS Code

1. **Open VS Code**
2. **Click on "File" → "Open Folder"**

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3. Select your **Gradle project folder**
4. VS Code will detect the Gradle project and **automatically import dependencies**.

---

4. Run the Gradle Java Project

Using VS Code Terminal

1. **Compile the project**:

$ gradle build

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2. **Run the application**:

sh

CopyEdit

gradle run

3. **Clean the build**:

sh

CopyEdit

gradle clean

Using VS Code GUI

1. Open the **Gradle Tasks** sidebar
2. Navigate to **Application → Run**
3. Click the **Run** button

---

5. Debugging the Project

1. Open **Main.java**
2. Add breakpoints
3. Click **Run → Start Debugging (F5)**
4. Select **"Java Debugger"**, and VS Code will start debugging.

What Type of Development is This?

This setup is specifically for **Java development** using **Gradle**.

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If you are looking for **Node.js development**, you would typically use:

1. **npm** or **yarn** instead of Gradle
2. **JavaScript/TypeScript** instead of Java
3. **Express.js, NestJS, or Fastify** frameworks

For **DevOps**, Gradle is often used in **CI/CD pipelines** (Jenkins, GitHub Actions, GitLab CI) to build and deploy Java applications.

Would you like a guide on integrating Gradle with DevOps tools? 🚀

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DevOps

DevOps is the buzzword you can't escape, and for good reason. It's a blend of Development and Operations—a methodology aimed at **streamlining the software development lifecycle (SDLC) by improving collaboration between developers and IT operations teams.

Think of DevOps as a recipe where:

1. Dev = Building the dish (coding)
2. Ops = Serving it piping hot to hungry users (deployment and maintenance)

Why Should You Care?

1. *Speed*: Faster delivery of features.
2. *Collaboration*: No more blaming each other—"It works on my machine" doesn't fly here.
3. *Reliability*: Fewer bugs in production (touch wood).
4. *Scalability*: Handle user traffic like a boss.
5. *Security*: Automate boring-but-important security checks.

The DevOps lifecycle involves a series of stages—Code, Build, Test, Configure, and Monitor—designed to streamline the software development process.

1. *Code*: Writing and managing code collaboratively.

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2. *Build*: Compiling source code into executable artifacts.
3. *Test*: Automating testing to ensure code quality.
4. *Configure*: Setting up the environment for deployment.
5. *Monitor*: Tracking application health in production.

Course objectives

To introduce DevOps terminology, definition & concepts

● To understand the different Version control tools like Git, Mercurial

● To understand the concepts of Continuous Integration/ Continuous Testing/ Continuous Deployment)

● To understand Configuration management using Ansible

● Illustrate the benefits and drive the adoption of cloud-based Devops tools to solve real world problems

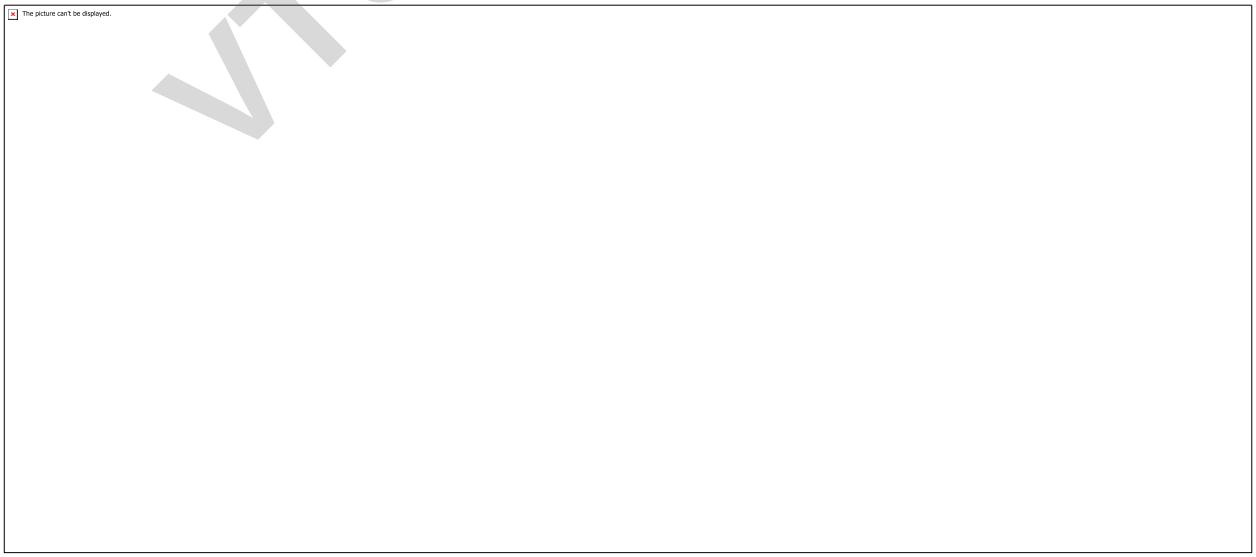
Course outcomes (Course Skill Set): At the end of the course the student will be able to:

● Demonstrate different actions performed through Version control tools like Git.

● Perform Continuous Integration and Continuous Testing and Continuous Deployment using Jenkins by building and automating test cases using Maven & Gradle.

● Experiment with configuration management using Ansible.

● Demonstrate Cloud-based DevOps tools using Azure DevOps.



Gradle

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Gradle is an open-source tool that helps us to create software with mechanization. This tool is widely used for the creation of different kinds of software due to its high performance. It works on Java and a Groovy-based Domain-Specific Language (DSL) for developing the project structure. Gradle supports the creation of mobile and web applications with testing and deploying on various platforms. With its functionality, it is preferred as an official tool for developing Android applications.

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Gradle is an open-source tool that helps us to create software with mechanization. This tool is widely used for the creation of different kinds of software due to its high performance. It works on Java and a Groovy-based Domain-Specific Language (DSL) for developing the project structure. Gradle supports the creation of mobile and web applications with testing and deploying on various platforms. With its functionality, it is preferred as an official tool for developing Android applications.

Advantages of Gradle:

1. Its highly customizable property. The tool can be modified under various technologies for diverse projects.
1. The performance of Gradle is very fast and efficient. It is around 2x in speed to that of Maven.
1. Gradle is a tool used for creating plug-ins and is a flexible instrument.
1. It provides a wide variety of IDEs for an enhanced user experience.

Disadvantages of Gradle:

1. Great technical expertise is required to build tasks with Gradle.
1. It does not come with inbuilt ant project structure.
1. Documentation of Gradle is somewhat extensive.
1. Ant build scripts are to be drafted with the help of XML. Moreover , to automate a difficult project , a lot of logic need to be written in XML files.

Maven

Maven is an open-source project management tool that helps us to create different software in the lifecycle used by this tool. This tool focuses on the standardization (i.e.) development of the software in a standard layout within a short duration of time. With this, we can create Java projects but is compatible to be used for other languages too. Maven uses Extensible Markup language(XML) for the structuring of the application.

Advantages of Maven:

1. The process of project building is simplified and well organized.

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1. Maven automatically executes the task of downloading Jar files and the other dependencies.
1. Maven can easily incorporate new dependencies by formulating the dependency code in the POM file.
1. It facilitates easy access to all the essential information.
1. It's extensible and plug-ins can be easily written using scripting languages or Java.

Disadvantages of Maven:

1. Installation in the working system is needed.
1. We can not implement a dependency using Maven if the Maven code for existing dependency is not found.
1. In terms of execution of project , Maven is quite slow.

Difference between Gradle and Maven

Basis	Gradle	Maven
Based on	Gradle is based on developing domain-specific language projects.	Maven is based on developing pure Java language-based software.
Configuration	It uses a Groovy-based Domain-specific language (DSL) for creating project structure.	It uses Extensible Markup Language (XML) for creating project structure.
Focuses on	Developing applications by adding new features to them.	Developing applications in a given time limit.
Performance	It performs better than maven as it optimized for tracking only current running task.	It does not create local temporary files during software creation and is hence – slower.
Java Compilation	It avoids compilation.	It is necessary to compile.

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Basis	Gradle	Maven
Usability	It is a new tool, which requires users to spend a lot of time to get used to it.	This tool is a known tool for many users and is easily available.
Customization	This tool is highly customizable as it supports a variety of IDE's.	This tool serves a limited number of developers and is not that customizable.
Languages supported	It supports software development in Java, C, C++, and Groovy.	It supports software development in Java, Scala, C#, and Ruby and it does not natively support C and C++ but can support through plugins like "maven-native-plugin" or we can integrate other build systems like CMake or Makefile.
Project Configuration	For declaring the project configuration, it does not use the XML files.	For declaring the project configuration, it uses the XML files.
Based on	Graph of task dependencies that do the work.	On the phases of the fixed and linear model.
Goal	To add functionality in the project is the main goal of the Gradle.	To finish the project in the given timeline is the main goal of the Maven.

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Source Code Management

Source Code Management is one of the key aspects of DevOps. It is the practice of tracking and managing the versions of your source code. Git is considered to be one of the best tools for

version control of source codes. It allows DevOps Engineers to collaborate, manage code, and implement CI/CD pipelines, code quality, and Infrastructure as a Code.

1. Introduction and Installation of Git

Git is a powerful and widely used version control system that helps developers track changes in their code, collaborate with others, and manage project history effectively. Whether you are a professional developer or just starting out, understanding Git is important for modern software development.

What is Git?

Git is an open-source distributed version control system created by Linus Torvalds in 2005. It allows multiple developers to work on a project simultaneously, without interfering with each other's changes. Git tracks and records changes in files, making it easier to collaborate, backtrack, and manage code efficiently.

Why Use Git?

1. **Collaboration**: Git enables multiple developers to work on the same project simultaneously. Changes can be merged seamlessly, and conflicts can be resolved easily.

1. **History Tracking**: Every change is recorded, allowing you to revert to previous versions of your code if something goes wrong.

1. **Branching and Merging**: Git allows you to create branches for new features or experiments without affecting the main codebase. Once the feature is ready, it can be merged back into the main branch.

1. **Distributed Development**: Each developer has a complete copy of the repository, including its history. This decentralization enhances collaboration and backup capabilities

Core Concepts of Git

1. Repositories

A **repository** (or repo) is a storage space where your project files and their history are kept. There are two types of repositories in Git:

1. **Local Repository**: A copy of the project on your local machine.

1. **Remote Repository**: A version of the project hosted on a server, often on platforms like GitHub, GitLab, or Bitbucket.

2. Commits

A **commit** is a snapshot of your project at a specific point in time. Each commit has a unique identifier (hash) and includes a message describing the changes made. Commits allow you to track and review the history of your project.

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3. Branches

A **branch** is a separate line of development. The default branch is called main or master. You can create new branches to work on features or fixes independently. Once the work is complete, the branch can be merged back into the main branch.

4. Merging

Merging is the process of integrating changes from one branch into another. It allows you to combine the work done in different branches and resolve any conflicts that arise.

5. Cloning

Cloning a repository means creating a local copy of a remote repository. This copy includes all files, branches, and commit history.

6. Pull and Push

1. **Pull**: Fetches updates from the remote repo
2.

LINK : https://github.com/manimozhisen/manimozhi_EPCET

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GitHub: Remote Repository

1. ttps://www.youtube.com/watch?v=Wx8_Rz1lKHg
2. How Git Version Control Works?
1. Useful Git Commands and Basic Concepts
1. Introduction to GitHub
1. List of useful GitHub Commands
1. Difference Between GitLab and GitHub
1. How to setup GitLab Repo in Windows 10
1. How to download and install GitLab?
1. Bitbucket vs GitHub vs GitLab

Step 0 : watch You tube link

Step 1: Sign up Git (https://github.com/) and create a Top repository through new and go to you specific path

Step 2 : install Git Bash (https://git-scm.com/downloads)

Step 3 :Open Git bash file in run mode or goto program file , from install Git bash to open it

Working directory → git add

Stagging area → Git Commit

Local repo → history will generate (once code / file is ready then do commit otherwise history will generate if you are not don't stagging area)

Remote repo

Step 4 : Run Git bash -- $ symbol will come

Step 5 : Create a dir or clone your remote repository

Step 6: to find out the git version

$ git --version

git version 2.47.1.windows.2

Step 7:$ ls (type ls command to get know the files)

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Step 8 : $ git init

The picture can't be displayed.

. is hidden file here of git

Step 9: $ ls -la

The picture can't be displayed.

Step 10 : mkdir students (this my working directory)

The picture can't be displayed.

Step 11: $ ls -la

The picture can't be displayed.

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Only . & .. only displaying.. so we should add git in epcet dir

Step 12 : create a git local repository thorugh git init command

The picture can't be displayed.

Step 13 :

Just checking git commit

The picture can't be displayed.

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Step 14 : create html file

$ vi p1.html

:wq! (press shift :wq! For quit html file)

The picture can't be displayed.

Step 15: Create a another text file

The picture can't be displayed.

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NOW P1 HTML FILE & 6TH SEM TEXT FILE

Step 16 : to find out the file under student dir (it’s a untrack file)

The picture can't be displayed.

Step 17:$ git status -s

The picture can't be displayed.

Step 18 : git add p1.html or git add .

The picture can't be displayed.

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Step 18:$ touch python

Step 19: $ touch java

Step 20 : $ ls

The picture can't be displayed.

Step 21: $ git status

The picture can't be displayed.

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2. Stacking fie are in green

3. Untracking files are in red color (java , python)

Step 22 : git add .

The picture can't be displayed.

Step 23 : $ git commit -s " this is Manimozhi work fr 1st program "

The picture can't be displayed.

Step 24 : all files will move to local repo if I have check git status

The picture can't be displayed.

Step 25 : git branch

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The picture can't be displayed.

NOW WE HAVE PUSH ALL FILE UNDER REMOTE REPOSITORY

Step 1 : go to you git webpage

Create a new repository under the name DEVOPS LAB PGMS

The picture can't be displayed.

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$ git add origin https://github.com/manimozhisen/DEVOPS-LAB-PGMS?tab=readme-ov-file#devops-lab-pgms

The picture can't be displayed.

$ git add origin

LINKS

Gradle

https://www.youtube.com/watch?v=I84f9Q5bFBA

https://github.com/techiescamp/devops-projects/tree/main/01-jenkins-setup

https://github.com/techiescamp/devops-projects

Maven

https://www.youtube.com/watch?v=rVLJjle2pjM

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```
$ java -version
$ download Gradle software / http://gradle.org / binary only
$ set the environment path in edit system environment / new
$ gradle -v
$ gradle init
$ gradle wrapper
$gradle bundle
$ gradlew.bat
```

Program 1 : Introduction to Maven and Gradle: Overview of Build Automation Tools, Key Differences Between Maven and Gradle, Installation and Setup

Download Gradle software:

Click download → gradle→ gardle.bat

Set edit system variable by C:\Users\Manimozhi Sen\Downloads\gradle-8.12.1-bin.zip\gradle-8.12.1\bin (path set up)

// students are instructed to set up path as per your work stations

C:\Users\Manimozhi Sen\Downloads\gradle-8.12.1-bin\gradle-8.12.1\bin>gradle -v

****

Maven

https://www.youtube.com/watch?v=sFfiYbyWX4I

Pre requisite

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Program:02

Experiment 2: Working with Maven: Creating a Maven Project, Understanding the POM File, Dependency Management and Plugins

1. Introduction to Maven

Maven is a powerful build automation and project management tool primarily used for Java projects. It simplifies the build process by:

- Enforcing a **standard project structure** (convention over configuration).
- Managing **dependencies** automatically by downloading them from remote repositories (e.g., Maven Central).
- Defining a clear **build lifecycle** (compile, test, package, install, deploy).
- Allowing the integration of various **plugins** to extend functionality (e.g., testing, reporting).

2. Creating a Maven Project

Step-by-Step Process

Step 1: Open Your Terminal

Make sure you have Maven installed (refer to Experiment 1). Open your terminal on your Ubuntu system.

Step 2: Use Maven Archetype to Generate a New Project

Maven comes with a set of archetypes that provide you with a standard project template. Use the following command to create a new Maven project:

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -
DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

- **groupId**: Uniquely identifies your project's group (like a package name).
- **artifactId**: The name of your project (the resulting artifact).

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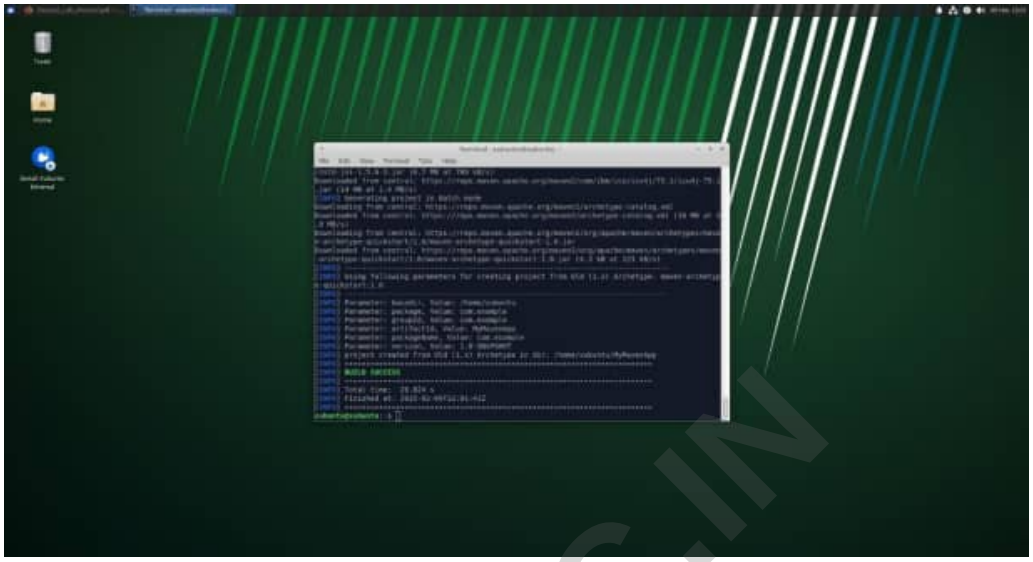
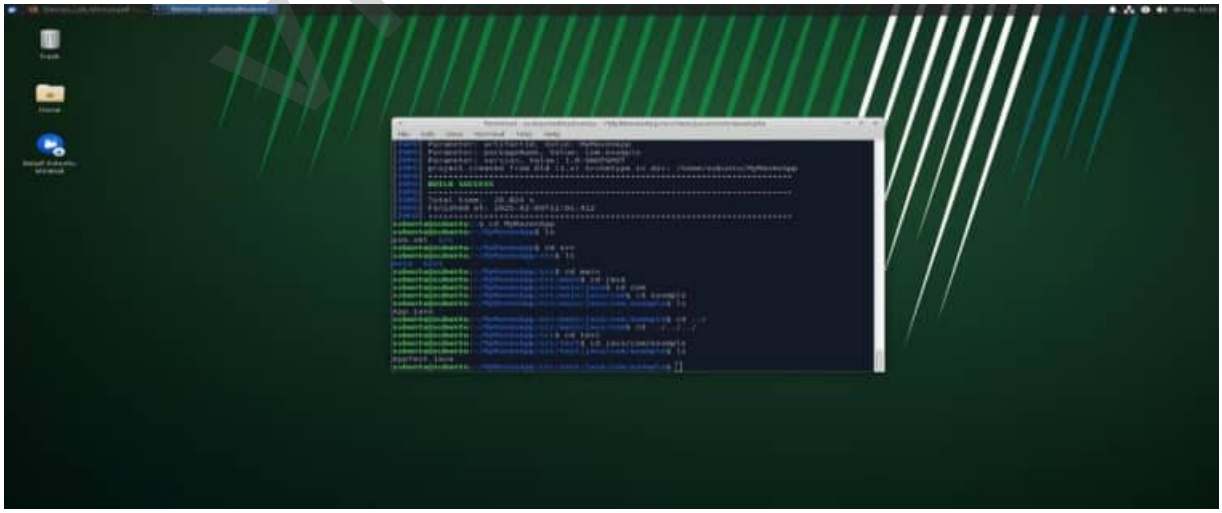
- **maven-archetype-quickstart**: A simple archetype that sets up a basic Java project with a sample unit test.
- **-DinteractiveMode=false**: Runs the command in non-interactive mode, using the provided parameters.



Step 3: Navigate to Your Project Directory

Once the command completes successfully, change your directory to the newly created project:

```
cd MyMavenApp
```



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3. Maven Project Layout and Components

After generating the project, you will notice the following standard Maven directory structure:

```
MyMavenApp/
├── pom.xml
└── src
    ├── main
    │   └── java
    │       └── com
    │           └── example
    │               └── App.java
    └── test
        └── java
            └── com
                └── example
                    └── AppTest.java
```

Explanation of Key Components:

- **pom.xml:**
  The **Project Object Model (POM)** file is the core of any Maven project. It contains configuration details such as project coordinates (groupId, artifactId, version), dependencies, plugins, and build settings.
- **src/main/java:**
  This directory holds the source code of your application. In our example, the package structure `com.example` is created, and you have an `App.java` file.
- **src/test/java:**
  This directory is for your test cases. The default example includes a basic test class, `AppTest.java`.

4. Understanding the POM File (pom.xml)

The `pom.xml` file is written in XML and is essential to how Maven operates. It includes several key sections:

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Basic Structure of a POM File

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
                             http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <!-- Project Coordinates -->
  <groupId>com.example</groupId>
  <artifactId>MyMavenApp</artifactId>
  <version>1.0-SNAPSHOT</version>

  <!-- Properties: Customize Java version or plugin versions -->
  <properties>
    <maven.compiler.source>11</maven.compiler.source>
    <maven.compiler.target>11</maven.compiler.target>
  </properties>

  <!-- Dependencies -->
  <dependencies>
    <!-- Example: JUnit dependency for testing -->
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <!-- Build: Configuring plugins and build settings -->
  <build>
    <plugins>
      <!-- Example: Maven Compiler Plugin to compile Java code -->
      <plugin>
```

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```
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>3.8.1</version>
        <configuration>
          <source>1.6</source>
          <target>1.6</target>
        </configuration>
      </plugin>

      <!-- Example: Maven Surefire Plugin to run tests -->
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-surefire-plugin</artifactId>
        <version>2.22.2</version>
      </plugin>
    </plugins>
  </build>
</project>
```

Key Elements Explained

- **Project Coordinates:**
  - `<groupId>`: Acts like a namespace, usually following the reverse domain name convention.
  - `<artifactId>`: The name of the project.
  - `<version>`: The current version of the project.
- **Properties:**
  - Used to define values that can be referenced elsewhere in the POM (e.g., Java source and target versions).
- **Dependencies:**
  - **Dependency Management:** Maven downloads and manages external libraries. In the example, JUnit is added for testing.
  - **Scope:** Determines when a dependency is used (e.g., `compile`, `test`, `runtime`).
- **Build and Plugins:**
  - **Maven Compiler Plugin:** Ensures that your Java code is compiled with the

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specified Java version.

- **Maven Surefire Plugin:** Executes unit tests during the build process.

5. Dependency Management with Maven

How Maven Manages Dependencies

- **Automatic Download:** When you specify a dependency in the `<dependencies>` section, Maven automatically downloads the library from a remote repository (usually Maven Central).
- **Transitive Dependencies:** Maven also resolves dependencies required by your dependencies.
- **Version Control:** You can specify precise versions for each dependency, ensuring consistency across builds.

Example Dependency

To add a dependency for JUnit, include the following snippet in your `<dependencies>` section:

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>4.13.2</version>
  <scope>test</scope>
</dependency>
```

- **groupId, artifactId, version:** These three elements uniquely identify the dependency.
- **scope:** The `test` scope ensures that this dependency is only available during the test phase and not included in the final artifact.

6. Maven Plugins: Extending Maven Functionality

Plugins are key to Maven's flexibility, adding tasks to your build process.

Example 1: Maven Compiler Plugin

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This plugin is used to compile your Java source code.

```
<plugin>
  <groupId>org.apache.maven.plugins</groupId>
  <artifactId>maven-compiler-plugin</artifactId>
  <version>3.8.1</version>
  <configuration>
    <source>11</source>
    <target>11</target>
  </configuration>
</plugin>
```

- **Configuration:** Specifies that the project should be compiled using Java 11.

Example 2: Maven Surefire Plugin

This plugin runs your unit tests during the `test` phase.

```
<plugin>
  <groupId>org.apache.maven.plugins</groupId>
  <artifactId>maven-surefire-plugin</artifactId>
  <version>2.22.2</version>
</plugin>
```

- **Purpose:** Automatically detects and runs tests in the `src/test/java` directory.

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7. Building and Testing Your Maven Project

Once your project is set up and your `pom.xml` is defined, you can use Maven commands to build and test your application.

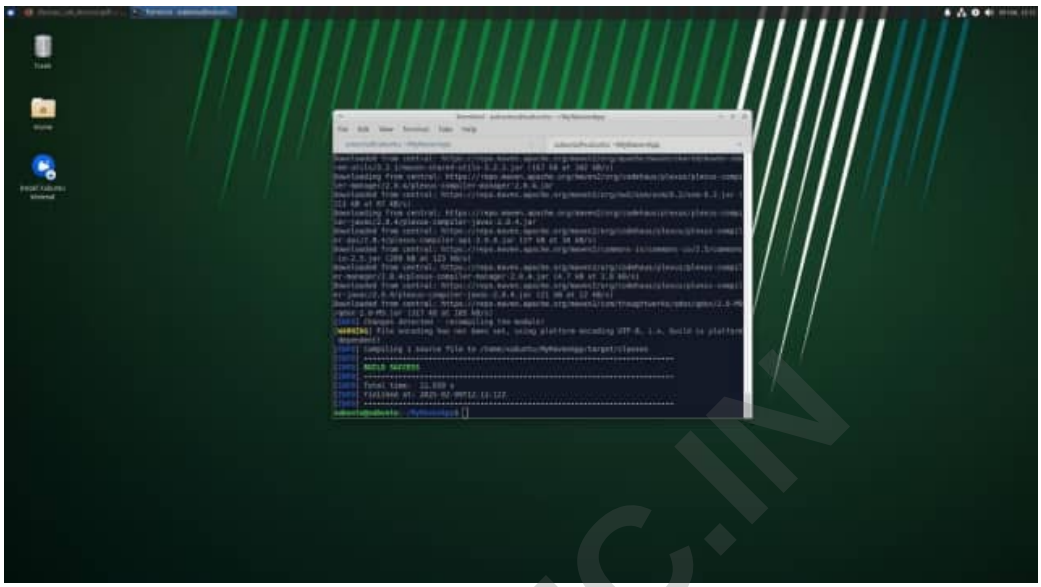
Common Maven Commands

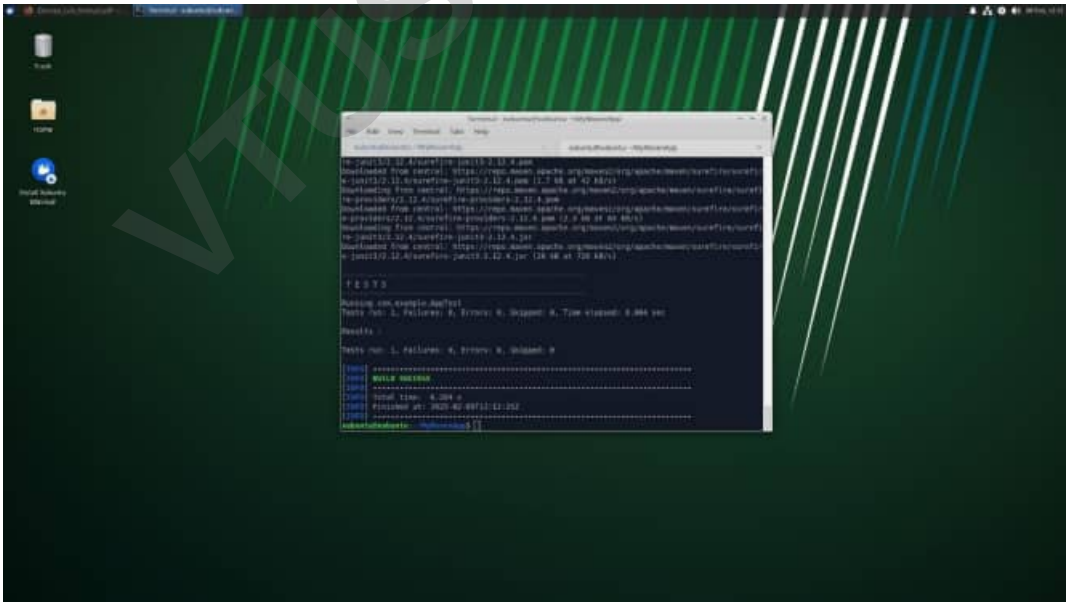
- **Compile the Project:**

```
mvn compile
```

- **Run Unit Tests:**

```
mvn test
```





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Package the Application:

```
mvn package
```

This command compiles, tests, and packages your code into a JAR file located in the `target` directory. *Screenshot Tip:* Capture the listing of the `target` directory showing the JAR file.

- **Clean the Project:**

```
mvn clean
```

This removes any files generated by previous builds.

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Experiment 3: Working with Gradle: Setting Up a Gradle Project, Understanding Build Scripts (Groovy and Kotlin DSL), Dependency Management and Task Automation

1. Introduction to Gradle

Gradle is a modern build automation tool designed to be highly flexible, fast, and scalable. It is widely used in Java projects, Android development, and many multi-language projects. Here's what makes Gradle stand out:

- **Flexible Build Scripts:** Gradle uses a Domain Specific Language (DSL) based on either Groovy (by default) or Kotlin. This provides a more dynamic and expressive way to define build logic compared to static XML configurations (as used in Maven).
- **Incremental Builds:** Gradle optimizes build times by determining what parts of the project have changed and rebuilding only those parts.
- **Task Automation:** Everything in Gradle is treated as a task, allowing you to create custom tasks or reuse existing ones for compiling code, running tests, packaging, and more.
- **Dependency Management:** Like Maven, Gradle can automatically download and manage dependencies from remote repositories (e.g., Maven Central, JCenter, or custom repositories).

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2. Setting Up a Gradle Project

Step-by-Step Instructions

Step 1: Ensure Gradle is Installed

Before starting, verify that Gradle is installed on your Ubuntu system. Open a terminal and run:

```
gradle -v
```

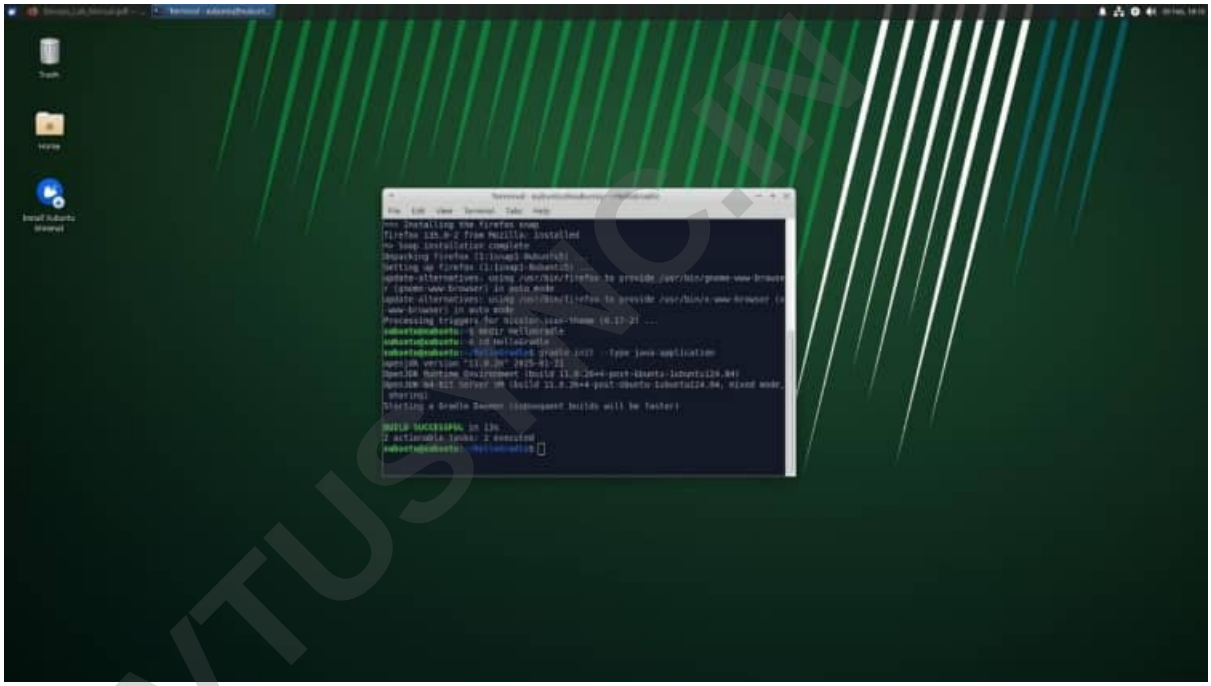
If you see version information, you're set. If not, follow the Gradle installation instructions from Experiment 1.

Step 2: Create a New Gradle Project

Gradle offers an interactive command to generate a new project using an initialization script. To create a basic Java application project:

1. **Create a new directory for your project and navigate to it:**
2. `mkdir HelloGradle`
3. `cd HelloGradle`
4. **Run the Gradle init command:**
1. `gradle init --type java-application`

You will be prompted to choose between a few options (if you don't use the non-interactive mode). Choose defaults or specify details as needed.



Step 3: Explore the Project Structure

After project creation, the directory structure typically looks like this:

```
HelloGradle/
├── build.gradle                // The primary build script (Groovy DSL by
default)
├── gradle/                  // Contains Gradle wrapper files (if generated)
├── gradlew                  // Unix shell script to run Gradle wrapper
├── gradlew.bat              // Windows batch script for Gradle wrapper
├── settings.gradle          // Contains project settings and names
```

```
└── src
```

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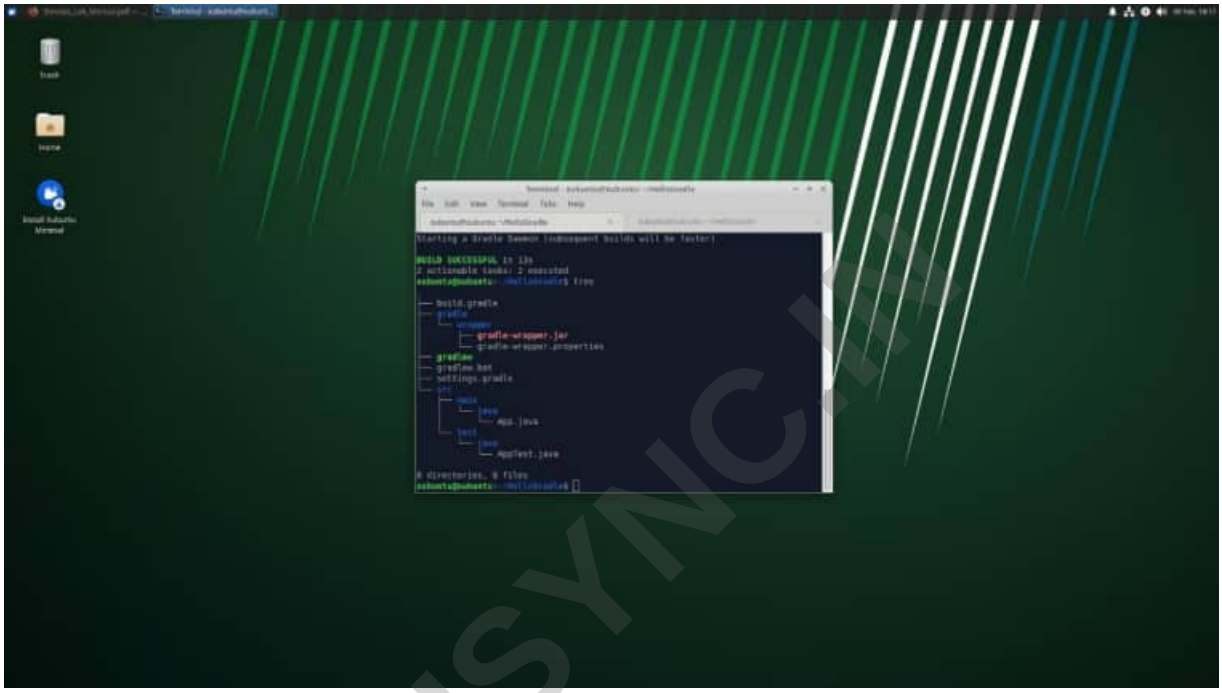
```
├── main
│   └── java
│       └── App.java     // Your main application source file
└── test
    └── java
        └── AppTest.java // Your test cases
```



Explanation of Components:

- **build.gradle:**
  This is the main build script written in Groovy (or Kotlin if you choose). It defines plugins, repositories, dependencies, and tasks.
- **settings.gradle:**
  A small script that defines the project's name and, in multi-project builds, the included subprojects.
- **gradlew / gradlew.bat:**
  The Gradle wrapper scripts. They allow you to run Gradle without requiring a separate installation on every machine by automatically downloading the correct Gradle version.
- **src/main/java:**
  Contains your application's source code.
- **src/test/java:**

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Contains your unit tests.

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3. Understanding Gradle Build Scripts

A **build script** in Gradle is a programmatic file that instructs Gradle on how to build your project. It can be written in two main DSLs:

- **Groovy DSL (build.gradle):** The traditional and most common syntax.
- **Kotlin DSL (build.gradle.kts):** A statically-typed alternative that leverages Kotlin's language features.

A. Groovy DSL Example (build.gradle)

Below is an example of a basic `build.gradle` for a Java application using Groovy DSL:

```
plugins {
    // Apply the Java plugin for compiling Java code
    id 'java'
    // Apply the application plugin to add support for building an application
    id 'application'
}

group = 'com.example'
version = '1.0'

repositories {
    // Use Maven Central for resolving dependencies.
    mavenCentral()
}

dependencies {
    // Define your dependencies. For example, JUnit for testing:
    testImplementation 'junit:junit:4.13.2'
}
```

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```
application {
    // Define the main class for the application.
    mainClass = 'com.example.App'
}
```

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```
// A custom task example: printing a greeting
task hello {
    doLast {
        println 'Hello, Gradle!'
    }
}
```

Explanation:

- **plugins block:** Declares the plugins used. The `java` plugin adds Java compilation and testing tasks, while the `application` plugin adds tasks for running the application.
- **group/version:** Sets project coordinates.
- **repositories block:** Configures the repository (Maven Central) where dependencies will be resolved.
- **dependencies block:** Lists the libraries your project depends on.
- **application block:** Specifies the main class of your application.
- **Custom task:** Defines a task named `hello` that prints a greeting when run.

B. Kotlin DSL Example (build.gradle.kts)

Here's how the same build configuration might look in Kotlin DSL:

Create a file named build.gradle.kts (or convert your file) with the following content:

```
plugins {
    // Apply the Java plugin for compiling Java code
    java
    // Apply the application plugin to add support for building an application
    application
}

group = "com.example"
version = "1.0"
```

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```
repositories {
    mavenCentral()

}
```

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```
dependencies {
    // Define dependencies using Kotlin DSL syntax
    testImplementation("junit:junit:4.13.2")
}

application {
    // Set the main class for the application
    mainClass.set("com.example.App")
}

// A custom task example using Kotlin DSL
tasks.register("hello") {
    doLast {
        println("Hello, Gradle with Kotlin DSL!")
    }
}
```

Explanation:

- **Kotlin DSL Syntax:** Uses a statically typed syntax, which can be more intuitive for developers familiar with Kotlin.
- **Plugins, Repositories, and Dependencies:** Defined similarly to the Groovy DSL but with Kotlin-style function calls and property access.
- **Custom Task Registration:** Uses `tasks.register` to define a task, similar in functionality to the Groovy DSL.

4. Dependency Management in Gradle

How It Works:

- **Repositories:** Gradle downloads dependencies from defined repositories (e.g., Maven Central).
- **Dependency Notation:** Dependencies are defined in a simple format: `group:artifact:version`.
- **Configuration Types:** Gradle provides various configurations such as

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`implementation`, `compileOnly`, `runtimeOnly`, and `testImplementation` to manage the scope of each dependency.

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Example:

In our build script examples, we included JUnit for testing:

```
dependencies {
    testImplementation 'junit:junit:4.13.2'
}
```

This line tells Gradle to download JUnit version 4.13.2 from Maven Central and include it in the test classpath.

Additional Dependency Configurations:

- **implementation:** Used for dependencies required at compile time.
- **runtimeOnly:** Used for dependencies required at runtime but not needed for compilation.

5. Task Automation in Gradle

In Gradle, nearly everything is a **task**. Tasks represent individual units of work (compiling code, running tests, packaging applications). Gradle comes with many built-in tasks (provided by plugins) and allows you to define your own custom tasks.

Built-in Tasks:

- **`compileJava`:** Compiles the source code in `src/main/java`.
- **`test`:** Runs tests in `src/test/java`.
- **`jar`:** Packages compiled code into a JAR file.
- **`run`:** Runs the application (if the application plugin is applied).

Creating Custom Tasks:

- **Custom Task in Groovy DSL:**

Ensure your `build.gradle` includes the following custom task:

```
groovy
CopyEdit
task hello {
```


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```
    doLast {
        println 'Hello, Gradle!'
    }
}
```

- **Run the Custom Task:**

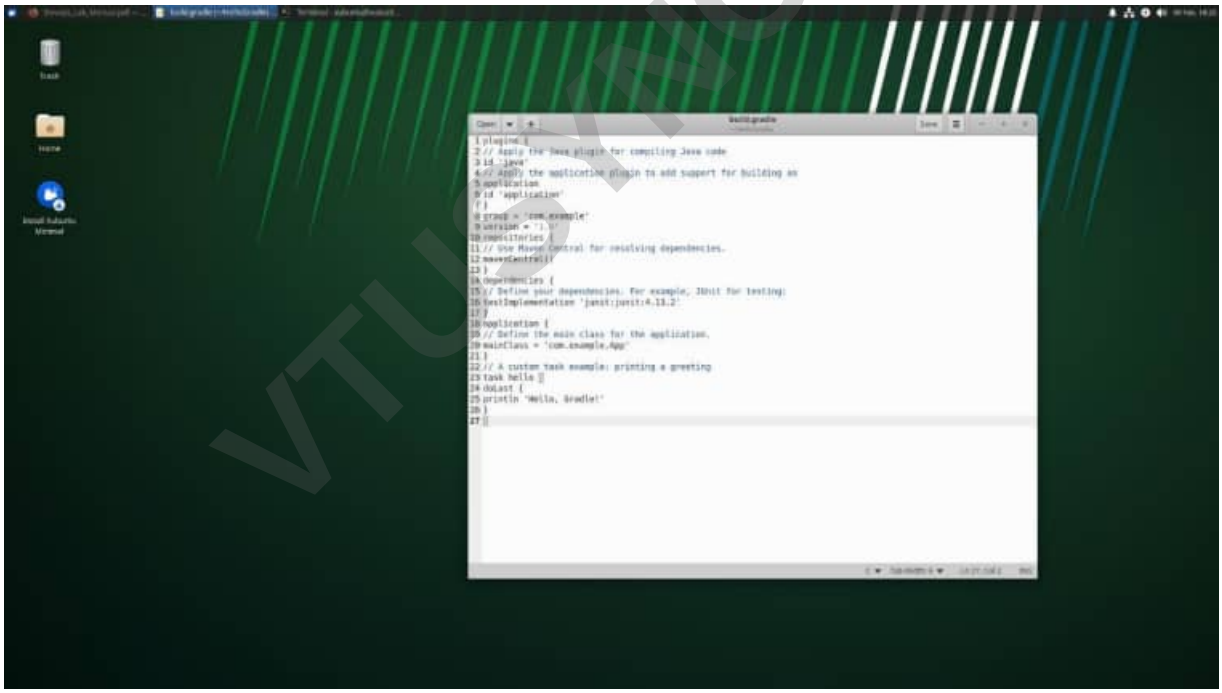
  - **Command:**

    ```
    gradle hello
    ```

  - **Expected Output:**

    You should see:

    ```
    > Task :hello
    Hello, Gradle!
    ```



- **Custom Task in Kotlin DSL (if using build.gradle.kts):**

Your Kotlin DSL file should include:

```
tasks.register("hello") {
    doLast {
        println("Hello, Gradle with Kotlin DSL!")
    }
```

```
}
```

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- **Run the Custom Task:**

  - **Command:**

    ```
    ./gradlew hello
    ```

    - **Expected Output:**

    You should see the greeting printed from the custom task.

Task Dependencies:

You can make one task depend on another. For instance, you might want your custom task to run after the `build` task:

Groovy DSL:

```
task greet(dependsOn: build) {
    doLast {
        println 'Build is complete! Time to celebrate!'
    }
}
```

Kotlin DSL:

```
tasks.register("greet") {
    dependsOn("build")
    doLast {
        println("Build is complete! Time to celebrate!")
    }
}
```

Screenshot Tip: Capture your terminal output after running these custom tasks.

6. Running and Verifying Your Gradle Project

Common Gradle Commands:

Build the Project:

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Compile, Test, and Package:

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- **Command:**

  ```
  gradle build
  ```

  - **What it does:**
    - Compiles source code (`compileJava`).
    - Runs tests (`test`).
    - Packages the application into a JAR file (`jar`).
  - **Expected Output:**

  Look for a "BUILD SUCCESSFUL" message.
- **Screenshot Tip:** Capture the full output of the `gradle build` command

Running the Application

Run the Application:

Command:

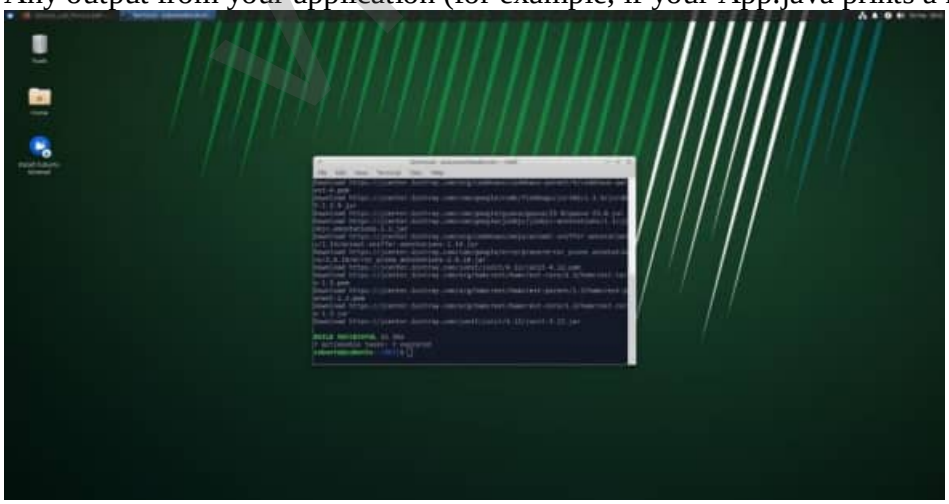
gradle run

What it does:

Runs your main class as specified in the application block.

Expected Output:

Any output from your application (for example, if your App.java prints a message).



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Executing Custom Tasks

Run the Custom Task:

Command:

```
gradle hello
```

Expected Output:

The custom greeting message you defined earlier.

Screenshot Tip: Capture the output showing the greeting.

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Experiment 4: Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle

Part A: Build and Run a Java Application with Maven

1. Create the Maven Project

1. **Open your Terminal.**
2. **Generate the Maven Project using the Quickstart Archetype:**

   Type the following command and press **Enter**:

   ```
   mvn archetype:generate -DgroupId=com.example -DartifactId=HelloMaven -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
   ```

   - **What it does:**
     This command creates a new Maven project with the group ID `com.example` and the artifact ID `HelloMaven`. The archetype sets up a basic Java application, including a sample test.
   - **Expected Output:**
     Maven will display messages as it downloads dependencies and generates the project files.
   - **Screenshot Tip:**
     Capture the terminal output showing the successful project generation.
3. **Change Directory into the Newly Created Project:**
4. `cd HelloMaven`
   - **Screenshot Tip:**
     Capture the output of the `pwd` or `ls` command to show the project structure.

2. Explore the Maven Project Structure

Your project directory should have a structure similar to this:

```
HelloMaven/
├── pom.xml
└── src
```


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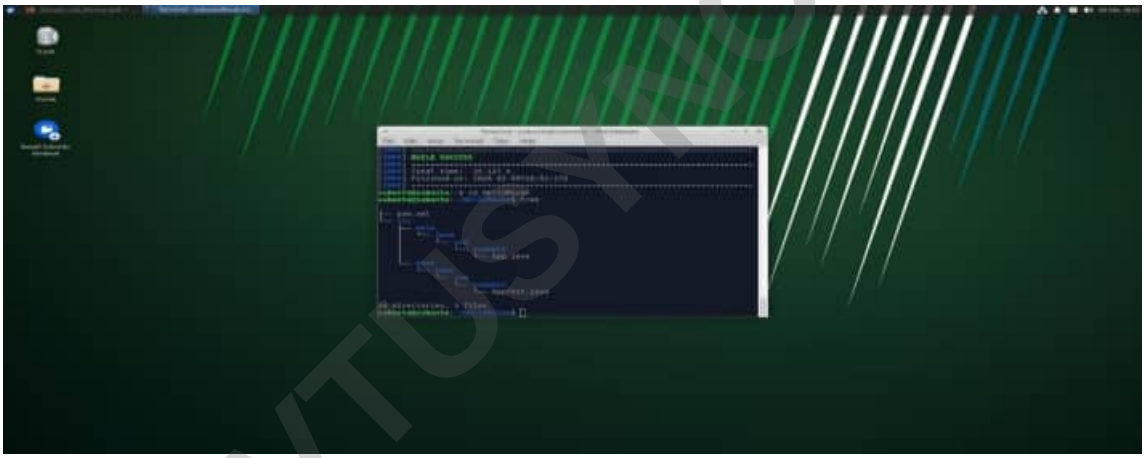
```
├── main
│   └── java
│       └── com
│           └── example
│               └── App.java
└── test
    └── java
        └── com
            └── example
                └── AppTest.java
```

- **`pom.xml`:** The Maven configuration file (POM) that defines your project's coordinates, dependencies, and plugins.
- **`src/main/java`:** Contains your application's source code.
- **`src/test/java`:** Contains unit tests.



3. Build the Maven Project

1. **Compile and Package the Application:**
2. `mvn package`
   - **What it does:** This command compiles the source code, runs tests, and packages your application into a JAR file (located in the `target` directory).

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- **Expected Output:**
  You should see a “BUILD SUCCESS” message along with information on the created JAR file.

4. Run the Maven Application

1. **Run the Application Using the JAR File:**

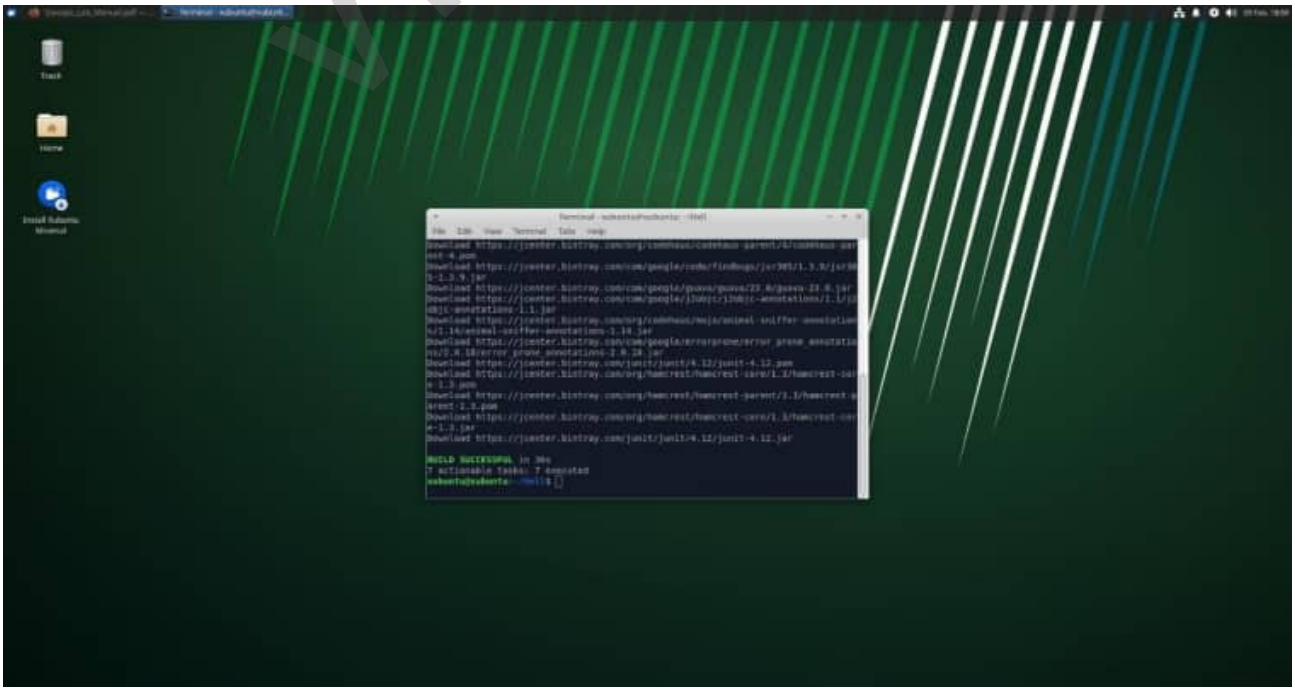
   Execute the following command:

```
java -cp target/HelloMaven-1.0-SNAPSHOT.jar com.example.App
```

- **What it does:**
  This command runs the `com.example.App` class from the JAR file generated in the previous step.
- **Expected Output:**
  The output should display:
- `Hello World!`

  (This is the default message printed by the generated `App.java`.)

- **Screenshot Tip:**
  Capture the terminal output showing “Hello World!” as proof that the application ran successfully.



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Part B: Migrate the Application to Gradle

In this part, you will create a Gradle project that contains the same Java application code, then build and run it using Gradle.

1. Create a New Gradle Project

1. **Open a New Terminal Window or Navigate Back to Your Workspace.**
2. **Create a New Directory for the Gradle Project:**
3. `mkdir HelloMavenGradle`
4. `cd HelloMavenGradle`
   - **What it does:**
     Creates and navigates into a folder named `HelloMavenGradle` for your new Gradle project.
5. **Initialize the Gradle Project Using the Java Application Type:**
6. `gradle init --type java-application`
   - **What it does:**
     Gradle will generate a basic Java application project with a default project structure.
   - **Expected Output:**
     You will see interactive prompts or a confirmation message stating that the project has been generated.

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2. Adjust the Gradle Project to Use the Same Code

The Gradle project structure will look similar to this:

```
HelloMavenGradle/
├── build.gradle
├── settings.gradle
└── src
    ├── main
    │   └── java
    │       └── App.java
    └── test
        └── java
```

```
└── AppTest.java
```

A. Replace or Update the Source Code:

1. **Copy the Application Code from the Maven Project:**
   - Open the Maven project directory (`HelloMaven/src/main/java/com/example/App.java`).
   - Copy its contents (which should print "Hello World!") into the Gradle project.
2. **Ensure the Package Declaration Matches:**
   - In the Maven project, the class is in the package `com.example`.
   - If the Gradle project's default `App.java` is not in this package, create the proper directory structure:
   - `mkdir -p src/main/java/com/example`
   - Move (or copy) the `App.java` file into the `com/example` folder:
   - `mv src/main/java/App.java src/main/java/com/example/`
   - Do the same for the test file if you wish to migrate tests.

B. Update the Gradle Build Script

1. **Open the `build.gradle` File in Your Editor:**
2. `nano build.gradle`
3. **Modify the `application` Block to Set the Correct Main Class:**

   Change or add the following in the file:

```
application {
    // Update the mainClass to reflect the package structure
    mainClass = 'com.example.App'

}
```

3. Build and Run the Gradle Application

1. **Build the Project:**
2. `gradle build`
   - **What it does:**
     Compiles the source code, runs tests, and packages the application.
   - **Expected Output:**
     Look for a "BUILD SUCCESSFUL" message.

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3. **Run the Application Using Gradle:**
4. `gradle run`
   - **What it does:**
     Uses the `application` plugin to run the main class defined in the `build.gradle` file.
   - **Expected Output:**
     The output should display:
   - `Hello World!`

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Experiment 5: Introduction to Jenkins: What is Jenkins?, Installing Jenkins on Local or Cloud Environment, Configuring Jenkins for First Use

1. What Is Jenkins?

Definition and Overview

Jenkins is an open-source automation server written in Java. It is widely used to facilitate **Continuous Integration (CI)** and **Continuous Deployment/Delivery (CD)** pipelines in software development. Jenkins automates repetitive tasks related to building, testing, and deploying software, helping teams to integrate changes continuously and deliver high-quality applications faster.

Key Functionalities

- **Continuous Integration/Delivery:** Automatically builds, tests, and deploys code after each commit.
- **Extensible Plugin Ecosystem:** Over 1,500 plugins allow integration with numerous tools such as Git, Maven, Gradle, Docker, and many cloud providers.
- **Automated Builds:** Jenkins can poll version control systems, trigger builds on code changes, and even schedule builds.
- **Pipeline as Code:** With the introduction of Jenkins Pipeline (using either scripted or declarative syntax), you can define your entire build process in a code file (Jenkinsfile) and store it alongside your code.
- **Distributed Builds:** Jenkins can distribute workloads across multiple machines, helping to run tests and builds faster.
- **Monitoring and Reporting:** Provides detailed logs, test reports, and build history.

Why Use Jenkins?

- **Improved Efficiency:** Automates mundane tasks to free up developer time.
- **Rapid Feedback:** Quickly identifies integration issues with automated tests.
- **Scalability:** Supports distributed builds and parallel testing.
- **Customization:** A highly configurable system that can integrate with nearly every tool in the software development lifecycle.

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- **Community Support:** Extensive community and plugin ecosystem that continuously evolve Jenkins' capabilities.

2. Installing Jenkins on a Local Machine (Ubuntu)

Below are detailed step-by-step instructions for installing Jenkins on an Ubuntu machine.

Step 1: Update Your System

Open your terminal and update your system repositories:

```
sudo apt update
sudo apt upgrade -y
```

Step 2: Install Java

Jenkins requires Java to run. It is recommended to use Java 11 or later. Install OpenJDK 11:

```
sudo apt install openjdk-11-jdk -y
java -version
```

Expected Output Example:

```
openjdk version "11.0.11" 2021-04-20
```

Step 3: Add the Jenkins Repository Key

Download and add the repository key so that your system trusts the Jenkins packages:

```
wget -q -O - https://pkg.jenkins.io/debian-stable/jenkins.io.key | sudo apt-key add -
```

Step 4: Add the Jenkins Repository

Add the Jenkins repository to your system's sources list:

```
sudo sh -c 'echo deb https://pkg.jenkins.io/debian-stable binary/ > /etc/apt/sources.list.d/jenkins.list'
```

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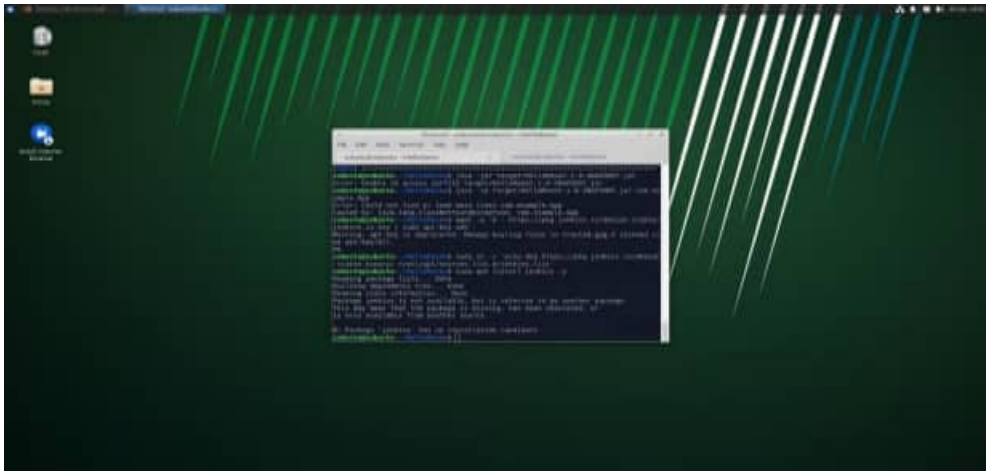

Step 5: Update the Repository and Install Jenkins

Update the package list and install Jenkins:

```
sudo apt update
sudo apt install jenkins -y
```

Step 6: Start and Enable the Jenkins Service

Start Jenkins and configure it to start automatically on boot:

```
sudo systemctl start jenkins
sudo systemctl status jenkins
```

- **What it does:**
  - `start jenkins` launches the Jenkins service.
  - `status jenkins` confirms Jenkins is running.
- *Expected Output:* A status message indicating Jenkins is active (running).

Step 7: Access Jenkins on the Local Machine

1. **Open a Web Browser:**
   Navigate to:
2. `http://localhost:8080`
3. **Unlock Jenkins:**
   The initial Jenkins screen will ask for an **administrator password**. Retrieve it by running:
4. `sudo cat /var/lib/jenkins/secrets/initialAdminPassword`

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- **What it does:** Displays the auto-generated admin password.
- *Screenshot Tip:* Capture the terminal output showing the password and the Jenkins unlock screen.

5. **Follow the Setup Wizard:**
    - **Install Suggested Plugins:** Click on "Install suggested plugins" for a typical setup.
    - **Create an Admin User:** Follow prompts to create your first admin user.
    - **Finalize Configuration:** Complete the remaining setup steps (e.g., instance configuration).
    - *Screenshot Tip:* Capture each major step (unlocking Jenkins, plugin installation, and admin user creation).

3. Installing Jenkins on the Cloud (Optional)

There are several ways to run Jenkins on the cloud. One common method is to run Jenkins using a Docker container on a cloud virtual machine. Below are instructions using Docker on an Ubuntu cloud server (for example, on AWS EC2, DigitalOcean, or any cloud provider that supports Ubuntu).

Step 1: Set Up a Cloud VM Running Ubuntu

1. **Provision a New Ubuntu Server:**
    - Create a new VM using your cloud provider's console.
    - Ensure the VM has at least 1–2 GB RAM.
    - Open port **8080** (and port **50000** for agent communication) in the security group settings.
    - Connect to the VM via SSH:
    - `ssh your-username@your-cloud-server-ip`

Step 2: Install Docker on Your Cloud VM

1. **Update the System:**
2. `sudo apt update`
3. `sudo apt upgrade -y`
4. **Install Docker:**
5. `sudo apt install docker.io -y`
6. `sudo systemctl start docker`
7. `sudo systemctl enable docker`

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8. **Verify Docker Installation:**
9. `docker --version`

Step 3: Run Jenkins in a Docker Container

1. **Pull the Official Jenkins Image:**
2. `sudo docker pull jenkins/jenkins:lts`
3. **Run the Jenkins Container:**
4. `sudo docker run -d --name jenkins -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts`
   - **What it does:**
     - `-d` runs the container in detached mode.
     - `--name jenkins` names the container.
     - `-p 8080:8080` maps port 8080 (Jenkins UI) to the host.
     - `-p 50000:50000` maps the agent communication port.
     - `-v jenkins_home:/var/jenkins_home` mounts a persistent volume for Jenkins data.
   - `sudo docker ps`
5. **Access Jenkins on the Cloud VM:**
   - Open your browser and navigate to:
   - `http://your-cloud-server-ip:8080`
   - Follow the same unlocking and setup wizard process as described in the local installation steps.

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4. Configuring Jenkins for First Use

After installing Jenkins (locally or on the cloud), complete these steps to configure it for first use:

Step 1: Unlock Jenkins

- **Retrieve the Admin Password:**
  As described earlier, run:
- `sudo cat /var/lib/jenkins/secrets/initialAdminPassword`
- **Enter the Password:**
  Copy the password into the "Administrator Password" field on the Jenkins unlock page.

Step 2: Install Suggested Plugins

- **Click "Install suggested plugins":**
  This option installs a set of commonly used plugins (e.g., Git, Maven Integration, Pipeline).
- **Wait for Installation:**
  The plugin installation process might take several minutes.

Step 3: Create Your First Admin User

- **Enter the Admin User Details:**
  Provide a username, password, full name, and email address.
- **Confirm Creation:**
  Click "Save and Continue" once the details are entered.

Step 4: Configure the Instance

- **Instance Configuration:**
  Jenkins may ask you to confirm the URL for your Jenkins instance. Verify that it is correct (e.g., `http://localhost:8080` for local installations or your cloud server's IP address for cloud installations).
- **Save the Configuration:**
  Confirm the settings and proceed.

Step 5: Start Using Jenkins

- **Dashboard:**
  Once the setup is complete, you will be taken to the Jenkins dashboard where you can:
  - Create new jobs (Freestyle projects, Pipelines, etc.)
  - Install additional plugins as needed.
  - Monitor builds and view logs.

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Experiment 6: Continuous Integration with Jenkins: Setting Up a CI Pipeline, Integrating Jenkins with Maven/Gradle, Running Automated Builds and Tests

1. Overview

What is a CI Pipeline?

A **Continuous Integration (CI) Pipeline** automates the process of building, testing, and integrating code changes every time code is committed to the repository. This pipeline:

- Automatically checks out the latest code.
- Compiles the application.
- Runs tests to catch errors early.
- Notifies the team of build/test results.

Why Use Jenkins for CI?

- **Automation:** Jenkins automates the build and test cycle, reducing manual intervention.
- **Immediate Feedback:** Developers get rapid notifications of any integration issues.
- **Extensibility:** With hundreds of plugins available, Jenkins can integrate with version control systems, build tools (Maven, Gradle), testing frameworks, and more.
- **Pipeline as Code:** Using Jenkins Pipelines (defined in a `Jenkinsfile`), you can manage the CI process as part of your source code repository.

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2. Setting Up a CI Pipeline with Jenkins (Freestyle Project)

This section explains how to create a CI pipeline as a Freestyle project that integrates with a Maven or Gradle project.

Step 1: Create a New Jenkins Job

1. **Log into Jenkins:**
    - Open your web browser and navigate to your Jenkins URL (e.g., `http://localhost:8080` or your cloud instance URL).
    - Log in with your admin credentials.

2. **Create a New Job:**
   - On the Jenkins dashboard, click on **"New Item"**.
   - **Enter an Item Name:** For example, `Maven-CI` (or `Gradle-CI` if you prefer Gradle).
   - **Select "Freestyle project"**.
   - Click **"OK"**.

Step 2: Configure Source Code Management (SCM)

1. **Select SCM:**
   - In the job configuration page, scroll down to the **"Source Code Management"** section.
   - Select **"Git"** (if using Git for version control).
2. **Enter Repository Details:**
   - **Repository URL:** Enter the URL of your Git repository (for example, `https://github.com/yourusername/your-maven-project.git`).
   - **Credentials:** If your repository is private, click **"Add"** to provide the necessary credentials.
   - Optionally, specify the **Branch Specifier** (e.g., `*/main`).

Step 3: Add Build Steps

A. For a Maven Project

1. **Add Maven Build Step:**
   - Scroll down to **"Build"** and click on **"Add build step"**.
   - Select **"Invoke top-level Maven targets"**.
   - **Goals:** In the Goals field, enter:
   - `clean package`

     This command instructs Maven to clean the previous build artifacts, compile the code, run tests, and package the application into a JAR/WAR file.

   - Optionally, set the **POM File** location if it is not in the default location (`pom.xml`).

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B. For a Gradle Project

1. **Add Gradle Build Step:**
   - If you are integrating a Gradle project, click on **"Add build step"** and choose **"Invoke Gradle script"** (this option might be available if you have installed a Gradle plugin in Jenkins).
   - **Tasks:** In the tasks field, enter:
   - `clean build`

     This instructs Gradle to clean previous build outputs and then build the project, running tests along the way.

   - **Switches:** If needed, you can add additional flags (for example, `--info` or `--stacktrace` for more detailed output).

Step 4: Configure Post-build Actions

1. **Publish Test Results:**
   - Scroll down to the **"Post-build Actions"** section.
   - Click **"Add post-build action"** and select **"Publish JUnit test result report"**.
   - **Test Report XMLs:** In the field, enter the pattern that matches your test report files. For example:
   - `**/target/surefire-reports/*.xml`

     for Maven, or a similar path for Gradle projects.

Step 5: Save and Run the Job

1. **Save the Configuration:**
   - Click **"Save"** at the bottom of the job configuration page.
2. **Trigger a Build:**
   - On the job's main page, click **"Build Now"**.
   - The build will be added to the build history on the left side.
3. **Monitor Build Output:**
   - Click on the build number (e.g., `#1`) and then click **"Console Output"**.

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- Verify that Jenkins successfully checks out the code, runs the build commands (Maven or Gradle), and executes tests.
- Look for **"BUILD SUCCESS"** or the equivalent output to confirm that the build and tests passed.

3. Setting Up a CI Pipeline with Jenkins (Pipeline as Code)

For greater flexibility and version-controlled CI configuration, you can use a **Jenkins Pipeline** defined in a `Jenkinsfile`.

Step 1: Create a Pipeline Job

1. **Log into Jenkins** and click on **"New Item"**.
2. **Enter an Item Name:** For example, `Pipeline-CI`.
3. **Select "Pipeline"** and click **"OK"**.

Step 2: Define the Pipeline Script

1. **Configure the Pipeline:**
   - In the job configuration page, scroll to the **"Pipeline"** section.
   - Choose **"Pipeline script"** (or "Pipeline script from SCM" if you want to load the script from your repository).
2. **Enter the Pipeline Script:**

   Below are sample pipeline scripts for Maven and Gradle projects.

   Example for a Maven Project:

```
pipeline {
    agent any

    stages {
        stage('Checkout') {
            steps {
                // Check out code from Git repository
                git url: 'https://github.com/yourusername/your-maven-
project.git', branch: 'main'
```

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```
                }
            }
            stage('Build') {
                steps {
                    // Run Maven build
                    sh 'mvn clean package'
                }
            }
            stage('Test') {
                steps {
                    // Optionally, separate test execution if needed
                    sh 'mvn test'
                }
            }
        }

        post {
            always {
                // Archive test reports
                junit '**/target/surefire-reports/*.xml'
            }
            success {
                echo 'Build and tests succeeded!'
            }
            failure {
                echo 'Build or tests failed.'
            }
        }
    }
```

Example for a Gradle Project:

```
pipeline {
    agent any
```

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```
    stages {
        stage('Checkout') {
            steps {
                // Check out code from Git repository
                git    url:    'https://github.com/yourusername/your-
gradle-project.git', branch: 'main'
            }
        }
        stage('Build') {
            steps {
                // Run Gradle build
                sh './gradlew clean build'
            }
        }
        stage('Test') {
            steps {
                // Run tests (if not already run in the build stage)
                sh './gradlew test'
            }
        }
    }

    post {
        always {
            // Archive test reports (modify the path according to your
project structure)
            junit '**/build/test-results/test/*.xml'
        }
        success {
            echo 'Build and tests succeeded!'
        }
        failure {
            echo 'Build or tests failed.'
        }
    }
```

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```
}
```

3. **Save the Pipeline Script:**
   - After entering your pipeline script, click **"Save"**.

Step 3: Run the Pipeline

1. **Trigger the Build:**
   - On the Pipeline job's main page, click **"Build Now"**.
   - Monitor the build progress through the Pipeline visualization or by clicking on the build number and then **"Console Output"**.
2. **Verify the Results:**
   - Confirm that each stage (Checkout, Build, Test) executes successfully.
- Review the archived test reports to verify that tests have run and passed.

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Experiment 7: Configuration Management with Ansible: Basics of Ansible: Inventory, Playbooks, and Modules, Automating Server Configurations with Playbooks, Hands-On: Writing and Running a Basic Playbook

1. Introduction to Ansible

What Is Ansible?

Ansible is an open-source IT automation and configuration management tool. It allows you to manage multiple servers and perform tasks such as:

- **Configuration Management:** Automate the configuration of servers.
- **Application Deployment:** Deploy applications consistently.
- **Orchestration:** Coordinate complex IT workflows and processes.

Key Concepts in Ansible

- **Inventory:**
  An inventory is a file (usually in INI or YAML format) that lists the hosts (or groups of hosts) you want to manage. It tells Ansible which machines to target.
- **Playbook:**
  A playbook is a YAML file that defines a set of tasks to be executed on your target hosts. It is the heart of Ansible automation. In a playbook, you specify:
  - **Hosts:** The target machines (or groups) on which the tasks should run.
  - **Tasks:** A list of actions (using modules) that should be executed.
  - **Modules:** Reusable, standalone scripts that perform specific actions (e.g., installing packages, copying files, configuring services).
- **Modules:**
  Ansible comes with a large collection of built-in modules (such as `apt`, `yum`, `copy`, `service`, etc.). These modules perform specific tasks on target hosts. You can also write custom modules if needed.

Why Use Ansible?

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- **Agentless:** Ansible uses SSH to communicate with target hosts, so no agent needs to be installed on them.
- **Simplicity:** Playbooks use simple YAML syntax, making them easy to write and understand.
- **Idempotence:** Ansible tasks are idempotent, meaning running the same playbook multiple times yields the same result, ensuring consistency.
- **Scalability:** Ansible can manage a small number of servers to large infrastructures with hundreds or thousands of nodes.

2. Installing Ansible on Ubuntu

Before writing a playbook, you need to install Ansible on your control machine (your local Ubuntu system).

Step 1: Update Your System

Open your terminal and run:

```
sudo apt update
sudo apt upgrade -y
```

Step 2: Install Ansible

Install Ansible using apt:

```
sudo apt install ansible -y
```

Verify the installation by checking the version:

```
ansible --version
```

Expected Output Example:

```
ansible 2.9.x
  config file = /etc/ansible/ansible.cfg
  ...
```

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3. Creating an Ansible Inventory

An inventory file lists the hosts you want to manage. For this experiment, you can use the local host.

Step 1: Create an Inventory File

1. Open your text editor to create a file called `hosts.ini`:
2. `nano hosts.ini`
3. Add the following content to define the local host:
4. `[local]`
5. `localhost ansible_connection=local`
    - **Explanation:**
        - `[local]` is a group name.
        - `localhost` is the target host.
        - `ansible_connection=local` tells Ansible to execute commands on the local machine without SSH.
6. Save the file by pressing **Ctrl+O** then **Enter**, and exit with **Ctrl+X**.

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3. Automated Server Configurations

While our experiment covered the basics, here's how you can extend it:

- **Configuring Services:** Use modules like `service` to start, stop, or restart services. For example, you can automate the configuration of web servers (e.g., Apache or Nginx).
- **Managing Files and Templates:** Use the `copy` or `template` modules to deploy configuration files across your servers. This is useful for maintaining consistent configuration settings.
- **User and Group Management:** The `user` and `group` modules allow you to create or modify user accounts, ensuring that the correct permissions and roles are applied automatically.

- **Advanced Orchestration:** Ansible playbooks can include conditionals, loops, and error handling to manage more complex setups, ensuring idempotence and consistency across your infrastructure.

Example: Automating a Web Server Configuration

If you wanted to automate the configuration of an Nginx server, your playbook might include tasks such as:

```
- name: Configure Nginx Web Server
  hosts: webservers
  become: yes
  tasks:

    - name: Update apt cache
      apt:
        update_cache: yes

    - name: Install Nginx
      apt:
        name: nginx
        state: present

    - name: Copy Nginx configuration file
      template:
        src: nginx.conf.j2
        dest: /etc/nginx/nginx.conf
      notify:
        - Restart Nginx

  handlers:
    - name: Restart Nginx
      service:
        name: nginx
        state: restarted
```

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5. Writing a Basic Ansible Playbook

You will now create a simple playbook that performs two common tasks:

- **Updating the apt cache**
- **Installing a package (e.g., `curl`)**

Step 1: Create the Playbook File

1. Open your text editor to create a file called `setup.yml`:

```
2. nano setup.yml
```

3. Add the following YAML content:

```
4. ---
5. - name: Basic Server Setup
6.   hosts: local
7.   become: yes  # Use privilege escalation (sudo)
8.   tasks:
9.     - name: Update apt cache
10.        apt:
11.          update_cache: yes
12.
13.      - name: Install curl
14.        apt:
15.          name: curl
16.          state: present
```

- **Explanation:**
  - `name`: Provides a descriptive name for the play.

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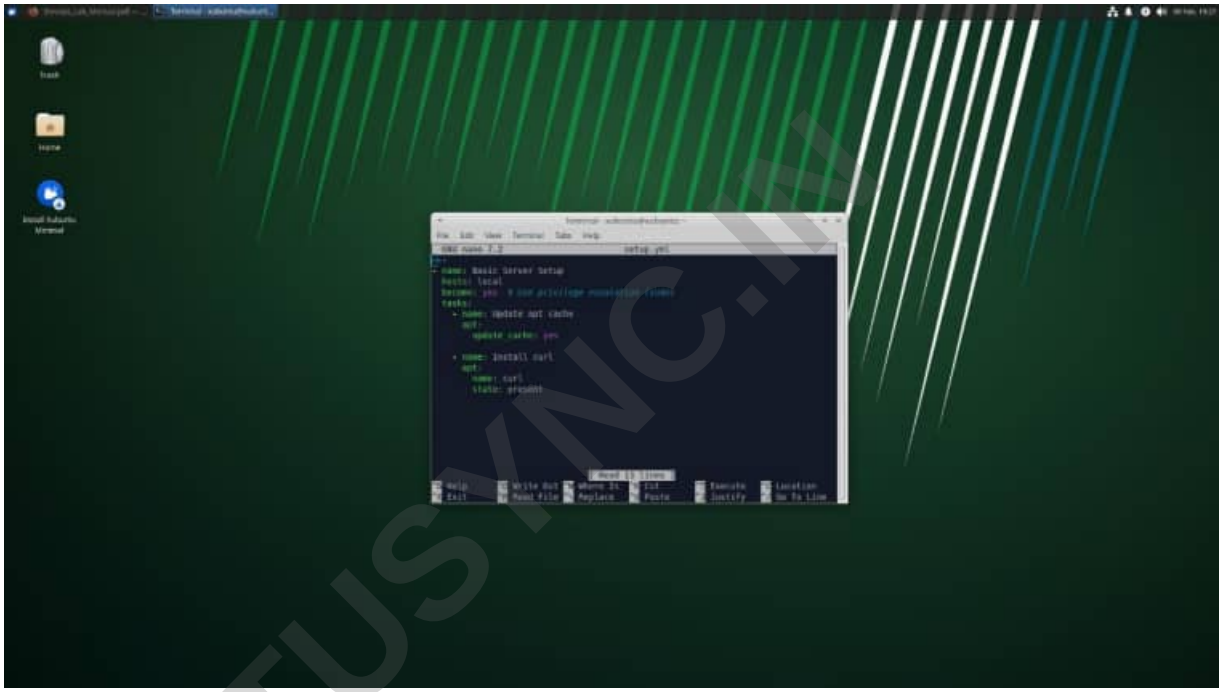
- `hosts`: Specifies the group or hosts from the inventory file.
- `become: yes`: Uses sudo to perform tasks that require elevated privileges.

- **Tasks Section:**
  - **Update apt cache:** Uses the `apt` module to update the package cache.
  - **Install curl:** Uses the `apt` module to install the `curl` package if it isn't already installed.

17. Save and exit the file.



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6. Running the Ansible Playbook

Step 1: Execute the Playbook

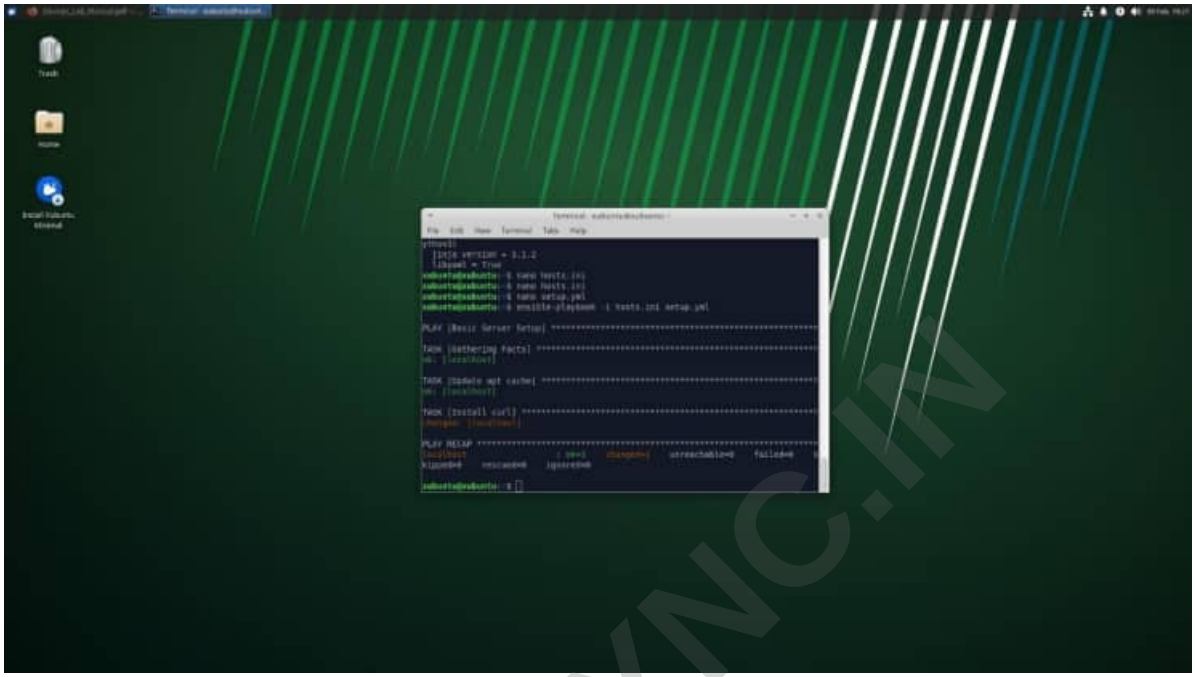
In your terminal, run the following command:

```
ansible-playbook -i hosts.ini setup.yml
```

- **Explanation:**
  - `ansible-playbook`: The command to run an Ansible playbook.
  - `-i hosts.ini`: Specifies the inventory file.
  - `setup.yml`: The playbook file you just created.

Expected Output:

- An output that details each task:
  - For example, a task summary might show “ok=2 changed=1” (if the apt cache was updated and curl was installed).



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Experiment 8: Practical Exercise: Set Up a Jenkins CI Pipeline for a Maven Project, Use Ansible to Deploy Artifacts Generated by Jenkins

1. Overview

In this experiment, you will:

- Set up a Jenkins job to automatically build a Maven project from source control.
- Archive the build artifact (a JAR file) produced by Maven.
- Integrate an Ansible deployment step within Jenkins (using a post-build action) to deploy the artifact to a target location.
- Verify that the artifact is deployed successfully.

This exercise demonstrates how Continuous Integration (CI) and automated configuration management can work together to streamline the build-and-deploy process.

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2. Prerequisites

Before you begin, ensure that:

- **Jenkins** is installed, running, and accessible (locally or on the cloud).
- **Maven Project:** You have a Maven project available in a Git repository (or stored locally). For this example, we will assume you are using the "HelloMaven" project generated in Experiment 2/4.
- **Git Repository:** The Maven project is committed to a Git repository (e.g., on GitHub) so Jenkins can pull the latest code.
- **Ansible Installed:** Ansible is installed on your control machine (or the Jenkins server) and you have created a basic inventory file (e.g., `hosts.ini`).

Tip: Verify installations and repository access before starting this exercise.

3. Step 1: Preparing the Maven Project

1. **Ensure Your Maven Project Is in Version Control:** If your "HelloMaven" project is not already in a Git repository, navigate to its root and initialize Git:

```
2. cd path/to/HelloMaven
3. git init
4. git add .
5. git commit -m "Initial commit of HelloMaven project"
```

Then push it to your repository (GitHub, GitLab, etc.).

Verify the Project Structure:

Your project should have a standard Maven layout:

```
6. HelloMaven/
7. ├── pom.xml
8. └── src
9.     ├── main/java/com/example/App.java
10.    └── test/java/com/example/AppTest.java
```

4. Step 2: Configuring Jenkins to Build the Maven Project

A. Create a New Jenkins Job (Freestyle Project)

1. **Log into Jenkins:**
   Open your browser and navigate to your Jenkins URL (e.g., `http://localhost:8080`).
2. **Create a New Job:**
   - Click **"New Item"** on the Jenkins dashboard.
   - **Enter an Item Name:** e.g., `HelloMaven-CI`.
   - **Select "Freestyle project"** and click **"OK"**.

B. Configure Source Code Management (SCM)

1. **Scroll to the "Source Code Management" Section:**
   - Select **"Git"**.
   - **Repository URL:** Enter your repository URL (e.g., `https://github.com/yourusername/HelloMaven.git`).
   - **Credentials:** If the repository is private, click **"Add"** and provide the necessary credentials.
   - **Branch Specifier:** (e.g., `*/main` or `*/master`).

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C. Add a Maven Build Step

1. **Scroll Down to the "Build" Section:**
   - Click **"Add build step"** and select **"Invoke top-level Maven targets"**.
2. **Configure the Maven Build:**
   - **Goals:** Type:
   - `clean package`

     This command cleans any previous builds, compiles the code, runs tests, and packages the application into a JAR file.

   - **POM File:** (Leave it as default if your `pom.xml` is in the root directory).

5. Step 3: Archiving the Artifact

After the Maven build completes, you need to archive the generated artifact so that it can be used later by the deployment process.

1. **Scroll Down to the "Post-build Actions" Section:**
   - Click **"Add post-build action"** and select **"Archive the artifacts"**.
2. **Configure Artifact Archiving:**
   - **Files to Archive:** Type:
   - `target/*.jar`

     This pattern tells Jenkins to archive any JAR file found in the `target` directory.

6. Step 4: Integrating Ansible Deployment in Jenkins

Now, integrate an Ansible deployment step into the Jenkins job. You can do this as a post-build action that executes a shell command.

1. **Add Another Post-build Action:**
   - Click **"Add post-build action"** and select **"Execute shell"**.
2. **Configure the Shell Command:**
   - In the command box, add a command to trigger your Ansible playbook. For example:
   - `ansible-playbook -i /path/to/hosts.ini /path/to/deploy.yml`

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Note:

- Replace `/path/to/hosts.ini` with the full path to your Ansible inventory file.
- Replace `/path/to/deploy.yml` with the full path to your Ansible deployment playbook.

- This command will run after a successful build, deploying the artifact using Ansible.

3. **Save the Jenkins Job:**
   - Click **"Save"** at the bottom of the configuration page.

7. Step 5: Writing an Ansible Playbook for Deployment

Create an Ansible playbook that deploys the Maven artifact (the JAR file) generated by Jenkins to a target directory.

A. Create an Inventory File

If you haven't already, create an inventory file (e.g., `hosts.ini`) that targets the deployment machine. For a local deployment, use:

```
[local]
localhost ansible_connection=local
```

B. Create the Deployment Playbook

1. **Open Your Text Editor** and create a file called `deploy.yml`:
2. `nano deploy.yml`
3. **Enter the Following YAML Content:**

```
---
- name: Deploy Maven Artifact
  hosts: local
  become: yes
  tasks:
    - name: Copy the artifact to the deployment directory
      copy:
src:      "/var/lib/jenkins/workspace/HelloMaven-CI/target/HelloMaven-1.0-SNAPSHOT.jar"
```

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```
12.          dest: "/opt/deployment/HelloMaven.jar"
```

Explanation:

- **hosts:** `local` means the playbook runs on the local machine. Adjust this if deploying to a remote server.
- **become: yes:** Uses sudo privileges to write to system directories.
- **src:** The path should point to the archived artifact in the Jenkins workspace. (Adjust the path if your Jenkins workspace is different.)
- **dest:** The target directory where you want the artifact deployed (ensure this directory exists or modify accordingly).

13. **Save and Exit the File**.

8. Step 6: Testing the Complete Pipeline

1. **Trigger a Build in Jenkins:**
   - Navigate to your Jenkins job (`HelloMaven-CI`) and click **"Build Now"**.
   - Monitor the build history. Once the build completes, click the build number (e.g., `#1`) and check the **Console Output**.
   - Look for messages indicating that:
     - The Maven build ran successfully.
     - The artifact was archived.
     - The shell command executed the Ansible playbook.

   Screenshot Tip: Capture the console output showing the full pipeline execution, including the deployment step.

2. **Verify Deployment:**
   - Log into your target machine (or check locally) and verify that the artifact has been copied to the destination directory (e.g., `/opt/deployment/HelloMaven.jar`).
   - For example, run:
   - `ls -l /opt/deployment/`
   - The output should list the deployed JAR file.

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Experiment 9: Introduction to Azure DevOps: Overview of Azure DevOps Services, Setting Up an Azure DevOps Account and Project

Overview of Azure DevOps

Azure DevOps is a comprehensive suite of cloud-based services designed to support the entire software development lifecycle. It provides tools for planning, developing, testing, delivering, and monitoring applications. Here are the primary services offered:

- **Azure Repos:**
  A set of version control tools that allow you to host Git repositories or use Team Foundation Version Control (TFVC). It offers collaboration features such as pull requests, branch policies, and code reviews.
- **Azure Pipelines:**
  A CI/CD service that helps automate builds, tests, and deployments. It supports multiple languages, platforms, and can run on Linux, Windows, or macOS agents.
- **Azure Boards:**
  A work tracking system that helps teams manage work items, sprints, backlogs, and Kanban boards. It facilitates agile planning and reporting.
- **Azure Test Plans:**
  Provides a solution for managing and executing tests, capturing data about defects, and tracking quality.
- **Azure Artifacts:**
  Allows you to create, host, and share packages (such as Maven, npm, NuGet, and Python packages) with your team, integrating package management into your CI/CD pipelines.

These services integrate with each other and with popular third-party tools to create a cohesive DevOps ecosystem.

Setting Up an Azure DevOps Account

Before you can start using Azure DevOps services, you need to set up an account and create an organization. Follow these steps:

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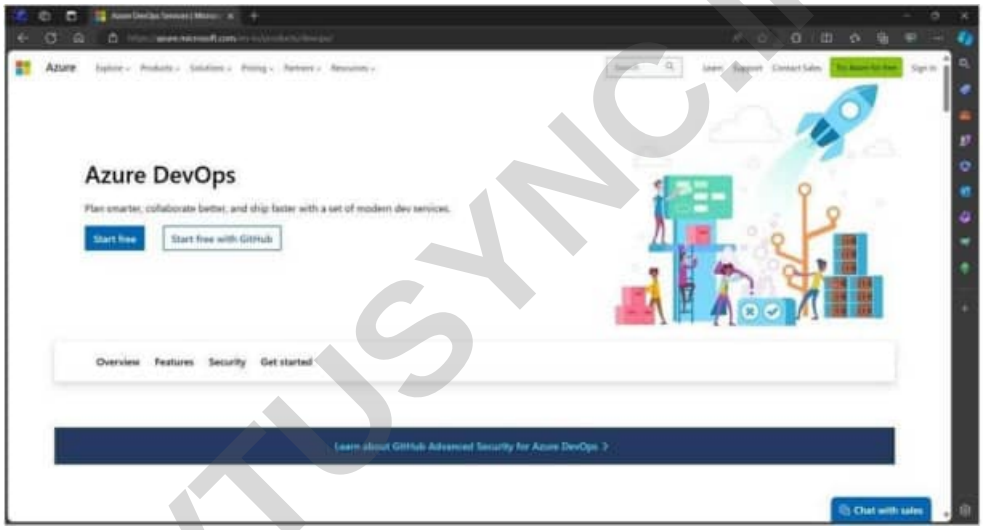
Step 1: Sign Up for an Azure DevOps Account

1. **Open Your Web Browser:**
   - Navigate to the Azure DevOps website: https://dev.azure.com.
2. **Sign In or Create a Microsoft Account:**
   - If you already have a Microsoft account (such as Outlook, Hotmail, or Office 365), click **"Sign in"**.
   - If you do not have a Microsoft account, click **"Create one!"** and follow the instructions to create a new Microsoft account.



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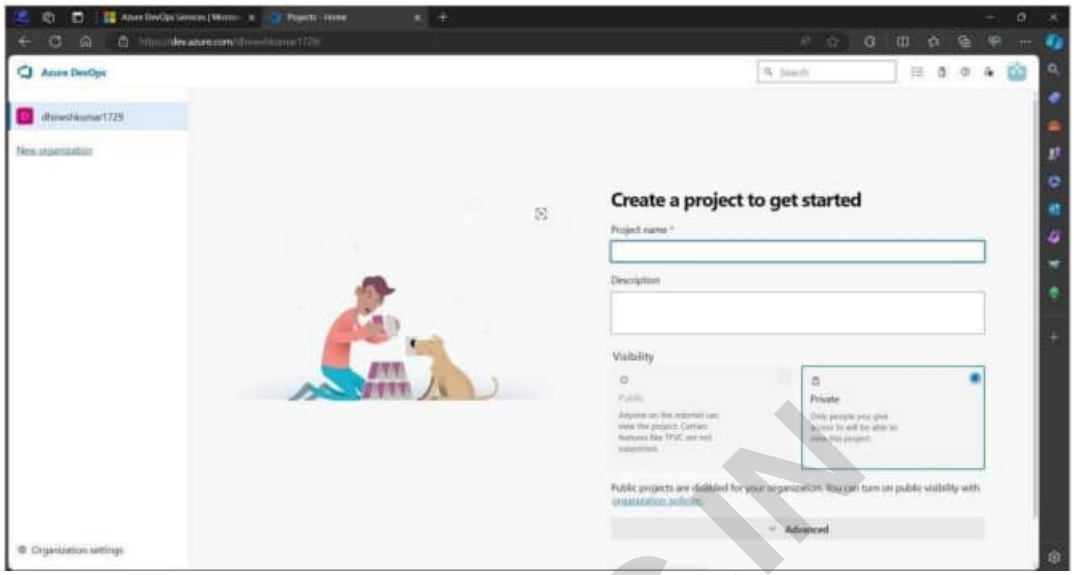
3. **Accept the Terms and Conditions:**
   - Review and accept the terms if prompted.

Step 2: Create an Azure DevOps Organization

1. **Create a New Organization:**
   - Once signed in, you will be prompted to create an Azure DevOps organization.
   - Enter a unique name for your organization (e.g., `YourCompanyDevOps` or `MyPersonalOrg`).

- **Select a Region:** Choose the geographic region where your data will be stored (select the one closest to you for optimal performance).
- Click **"Continue"** or **"Create"**.



2. **Review Your Organization's Dashboard:**
   - Once created, you will see an overview dashboard for your organization. This dashboard provides navigation links to Repos, Pipelines, Boards, Test Plans, and Artifacts.

3. Creating an Azure DevOps Project

After setting up your organization, the next step is to create a project. A project in Azure DevOps is a container for all your source code, pipelines, boards, and other resources.

Step 1: Create a New Project

1. **Navigate to "New Project":**
   - On your organization's dashboard, click the **"New Project"** button.
2. **Configure Your Project:**
   - **Project Name:** Enter a descriptive name for your project (e.g., `HelloDevOps`).
   - **Description:** Optionally, provide a brief description (e.g., "A sample project to demonstrate Azure DevOps services").
   - **Visibility:**

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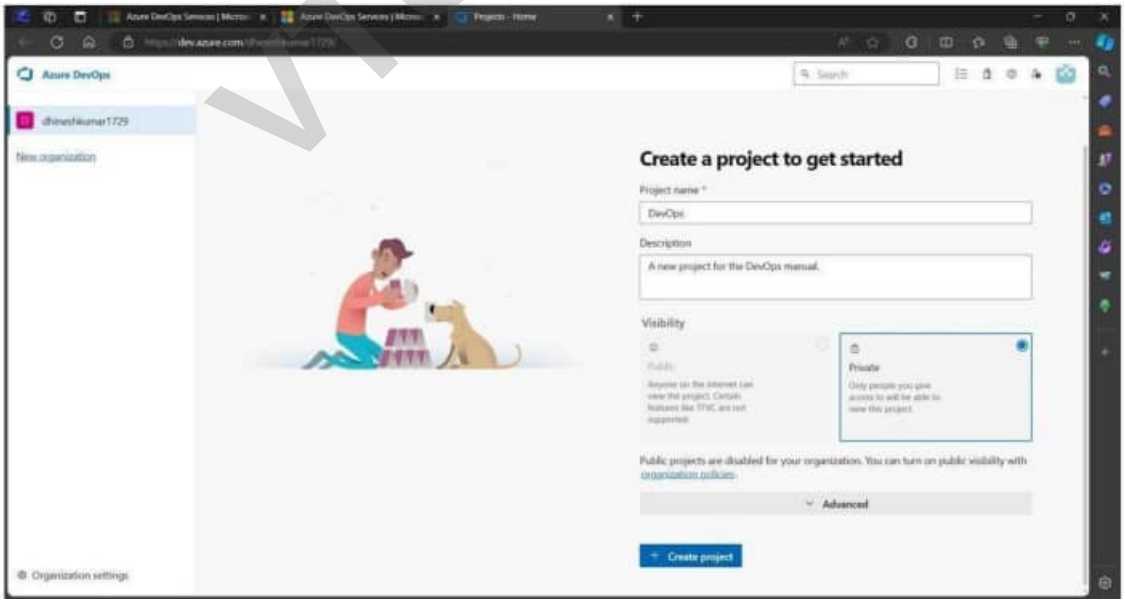
- Choose **"Private"** if you want to restrict access to your project.
- Choose **"Public"** if you are okay with the project being accessible to anyone.

- **Advanced Options (Optional):** You can choose a version control system (Git is the default) and a work item process (Agile, Scrum, or Basic). For most beginners, the defaults are recommended.
- Click **"Create"**.

Step 2: Explore Your Project Dashboard

1. **Project Overview:**
   - Once your project is created, you will be directed to the project dashboard. Here you will see navigation options for:
     - **Repos:** Where your code is stored.
     - **Pipelines:** For build and release automation.
     - **Boards:** For work tracking and agile planning.
     - **Test Plans:** For managing and running tests.
     - **Artifacts:** For hosting packages.
2. **Familiarize Yourself with the Interface:**
   - Click through each section (e.g., Repos, Pipelines, Boards) to get a sense of the available features.



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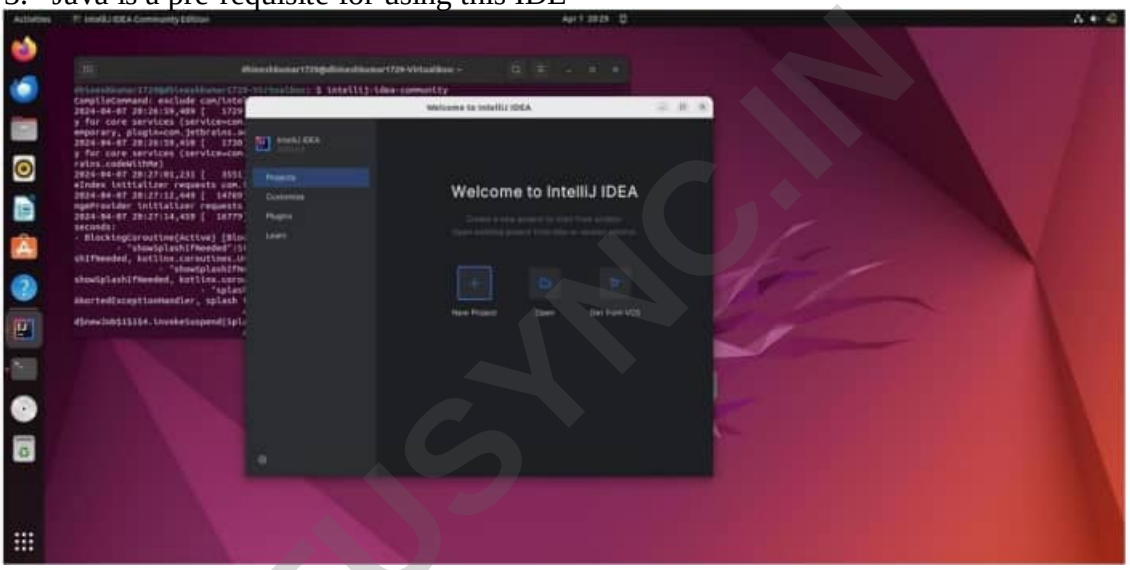
Experiment 10: Creating Build Pipelines: Building a Maven/Gradle Project with Azure Pipelines, Integrating Code Repositories (e.g., GitHub, Azure Repos), Running Unit Tests and Generating Reports

Step 1: Installing IntelliJ IDEA

1. Download IntelliJ Idea from https://www.jetbrains.com/idea/download/?section=linux and activate a free trial for 30 days or through the following command. $ sudo snap install intellij-idea-community --community
2. Open the IDE using the below command in the terminal. $ intellij-idea-community
3. Java is a pre-requisite for using this IDE



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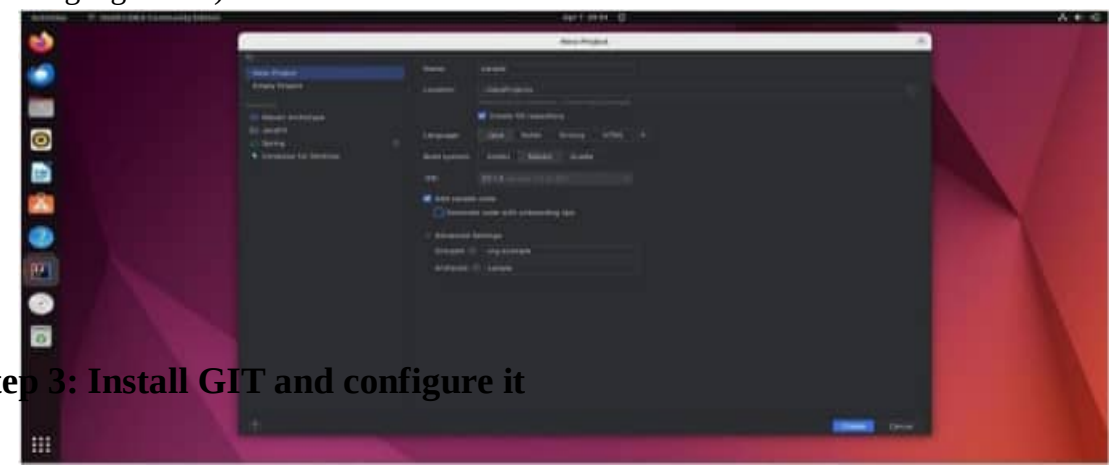
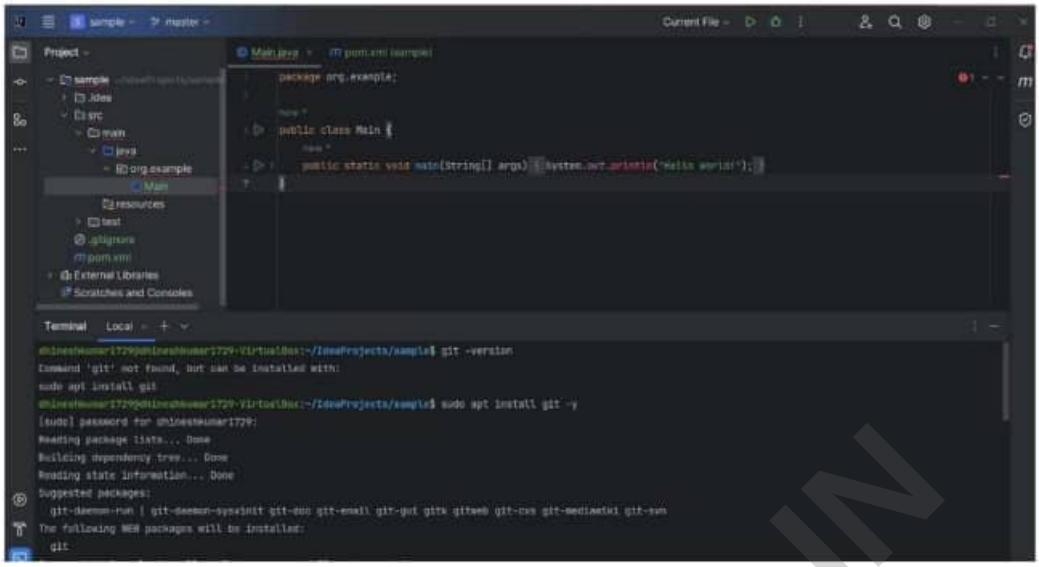
Step 2: Create a new Maven project using the IDE.

Create a project by clicking on the New Project option, choose an appropriate name and select the following features as mentioned in the image (Build system: Maven andLanguage: Java)



Step 3: Install GIT and configure it

1. Check whether git is installed using the following command. $ git -version
2. Install git using $ sudo apt install git
3. Once git has been installed execute the following commands one by one.



$ git init $

git add .

$ git commit -m “First Commit”

$ git branch -M main

Configure git using the below commands

$ git config --global user.name “Your_Name”

$ git config --global user.email “Your_Email_ID”

Go to https://github.com sign in with your account and create a new private repository with the same name as your maven project.

1. Copy the SSH URL for the created repository.

2. Now go back to IDE’s terminal and execute the following commands

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$ ssh-keygen -t rsa -b 4096 -C youremail@gmail.com

(press ENTER for all questions)

$ cat ~/.ssh/id_rsa.pub (copy the printed SSH key)

1. Now go to the SSH and GPG section in the GitHub settings option.

2. Click on the New SSH key button.

3. Choose a suitable name and paste the SSH key into the provided space.

4. Now get back to the IDE's terminal and type the following command

$ git remote set-url origin git@github.com:repo_owner/repo_name.git

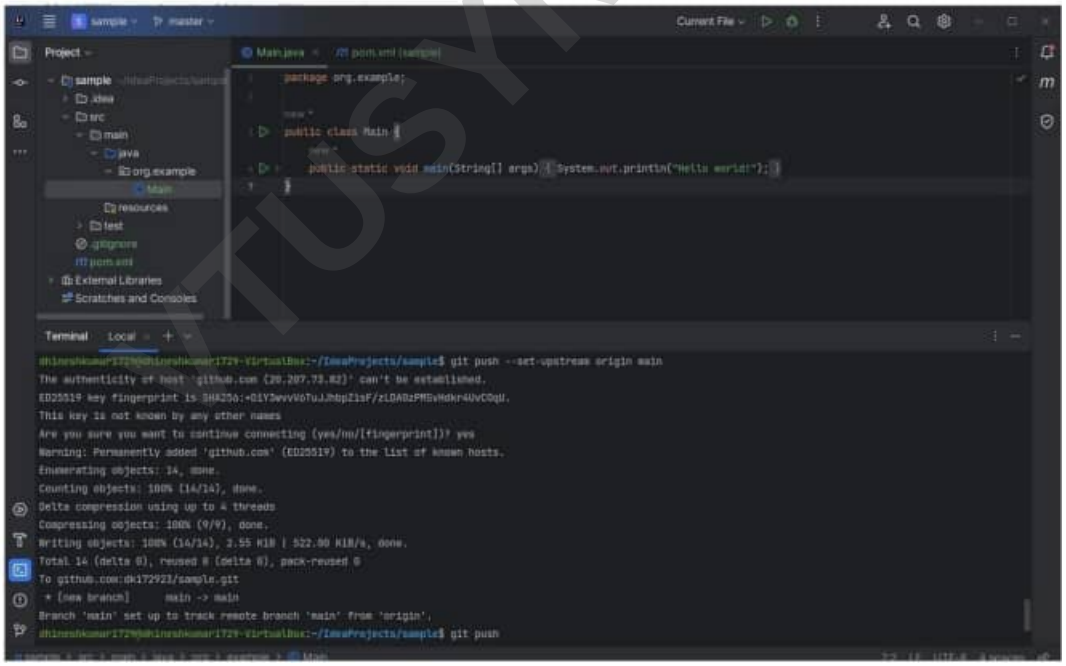
(Here, git@github.com:dk172923/sample.git is the remote URL to use SSH

dk172923 – GitHub ID sample.git – repository name)

Now finally run the push commands to push the code to the remote repository from the local repository.

$ git push --set-upstream origin main (Type YES for prompted question)
$ git push

$ git push



The code has been successfully pushed to the remote repository.

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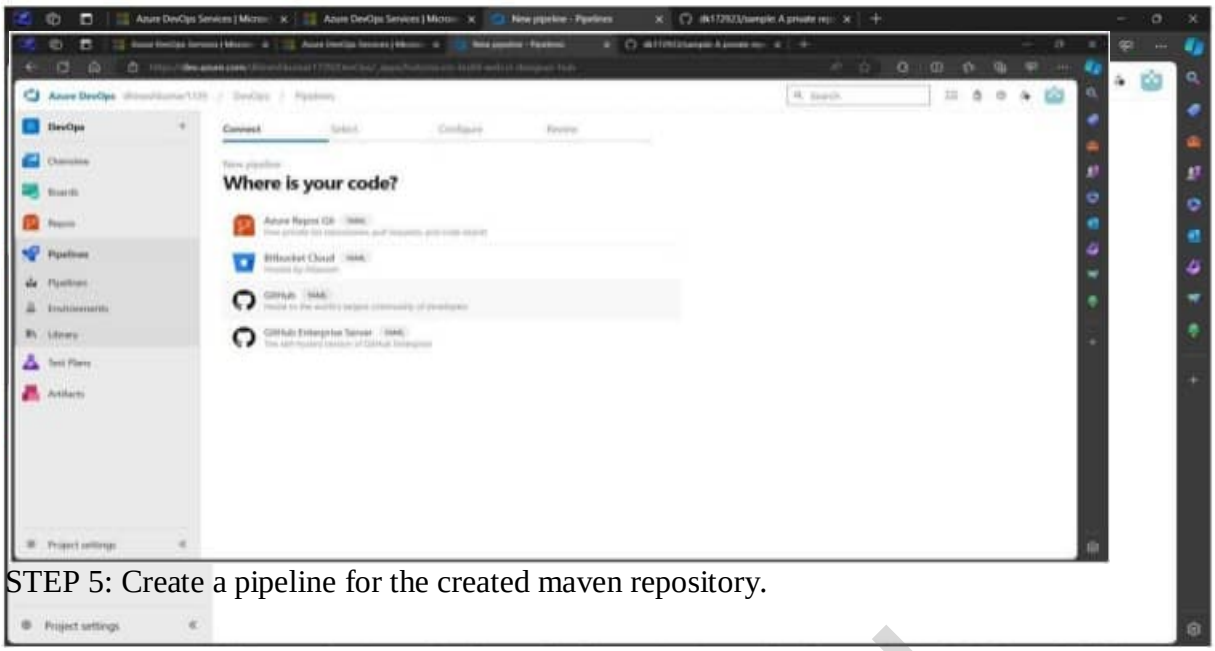
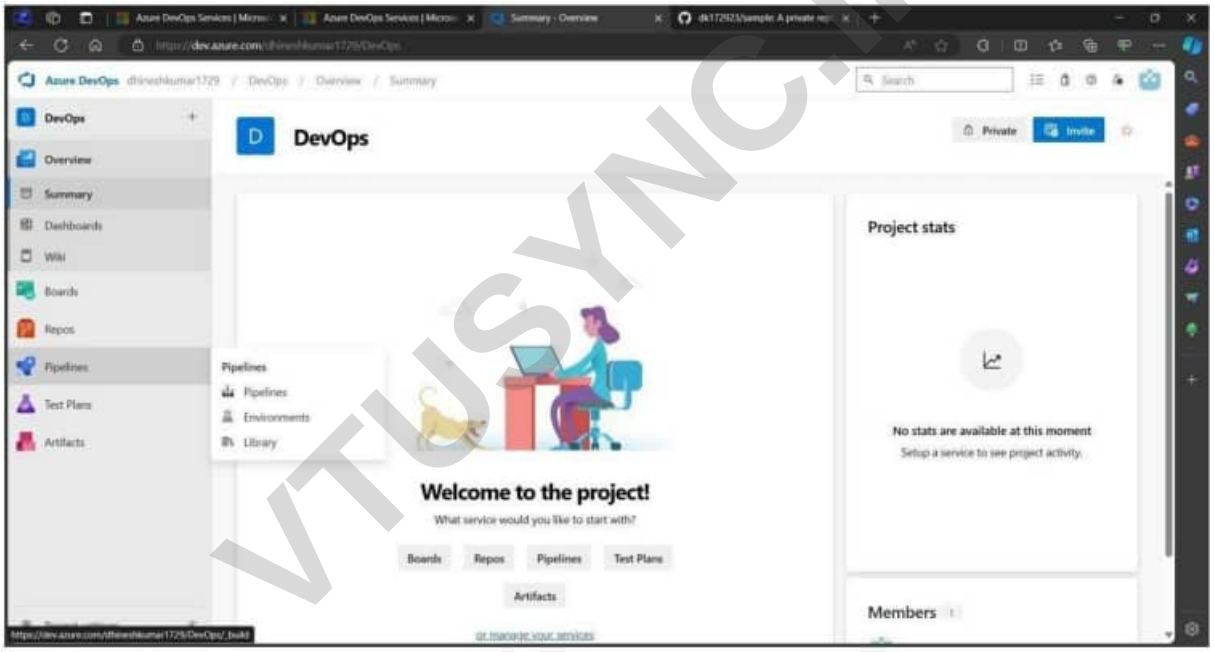


STEP 5: Create a pipeline for the created maven repository.

Click on Pipelines on the left pane of the Azure DevOps website.



1. Click on the Create Pipeline button.

2. Choose the GitHub YAML option.

1. Select the created repository “sample” from the available repositories.

2. Give permissions if prompted to do so and sign in to your GitHub account if needed.

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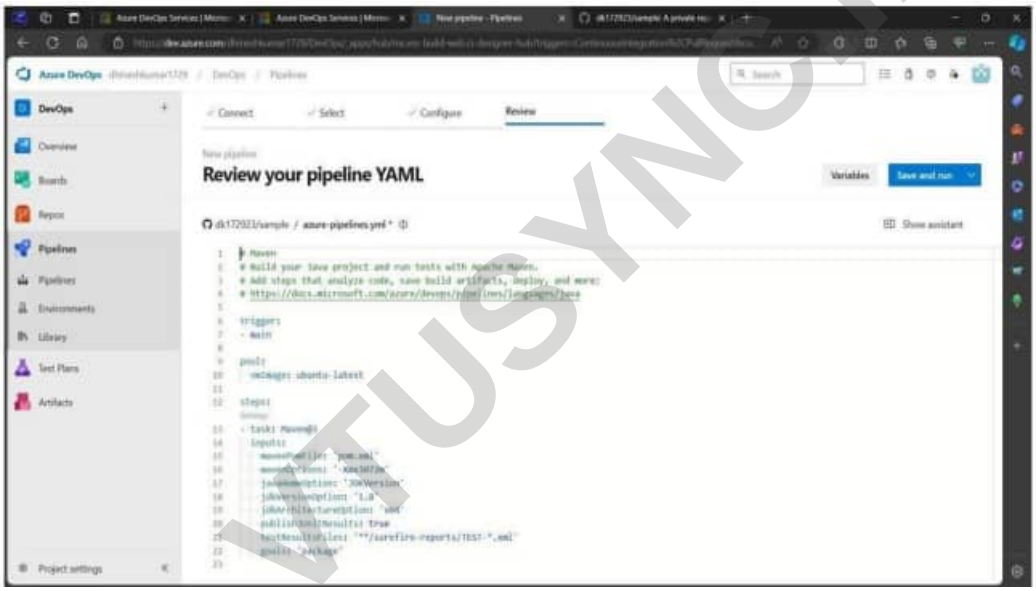
3. Choose Maven Pipeline from the given options.

A YAML file is automatically created based on the configuration details found in the

4. pom.xml file in the repository.



1. Create a commit message and press the run button again.

2. Click on the created Job from the Jobs table.

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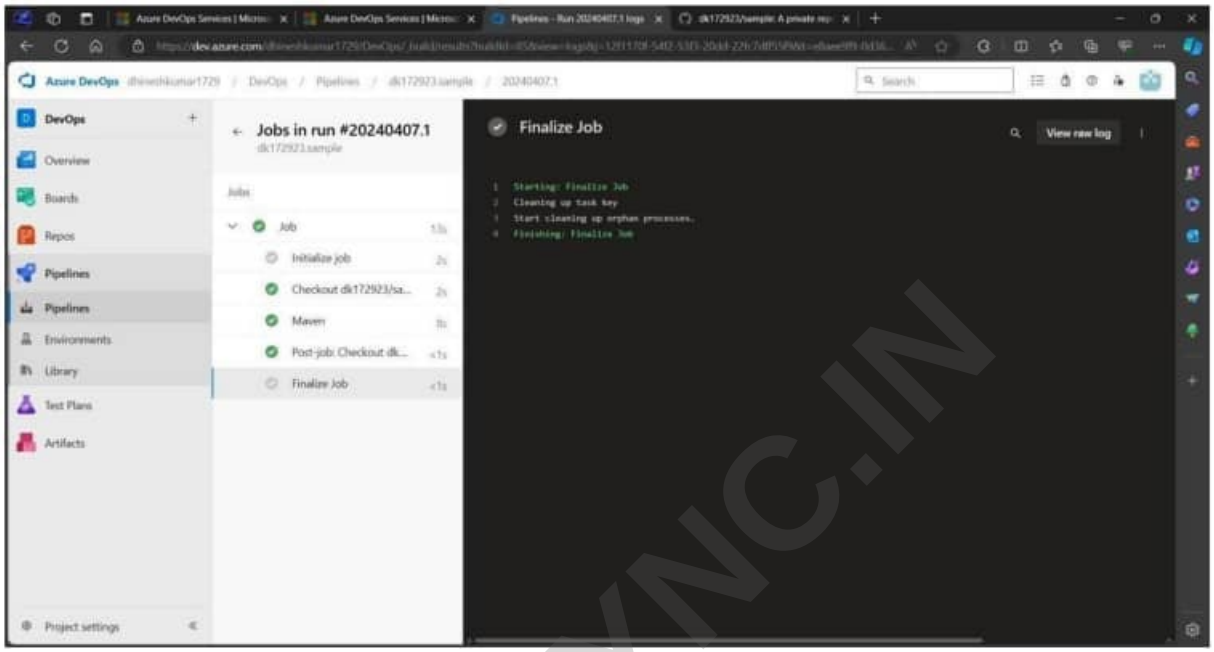
A successful job completion would be like this.



Below is a detailed section that explains how to run unit tests and generate reports in an Azure DevOps pipeline after your Maven project has been built. This section explains what happens during the Maven build process, how test results are generated, and how to publish these results as part of your pipeline execution.

Running Unit Tests and Generating Reports with Maven in Azure DevOps

When you build your Maven project using Azure Pipelines, the build process usually includes running unit tests with the Maven Surefire plugin. This plugin executes tests (typically written with JUnit) and produces test result files in XML format. Azure Pipelines can then pick up these XML files and present them as part of the build summary. Below are the steps and details to ensure that your unit tests are executed and the reports are published.

1. **Maven Surefire Plugin and Test Reports What Happens During the Maven Build?**

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- **Maven Surefire Plugin:** When you run the command mvn clean package (or mvn test), the Surefire plugin automatically executes the unit tests found in the src/test/java directory.
- **Test Report Generation:** By default, the Surefire plugin creates XML reports in the target/surefire-reports/ directory. These files usually have names like TEST-<TestClassName>.xml.
- **Importance of Test Reports:** These XML files contain detailed information on test execution, including the number of tests run, passed, failed, and any error messages or stack traces.

2. Configuring Your Azure Pipeline to Publish Test Results After your Maven build runs and tests are executed, you need to add a step in your Azure Pipeline YAML file that locates these test reports and publishes them in Azure DevOps. This is accomplished by using the **PublishTestResults@2** task.

YAML Configuration Example for Maven

Below is a sample snippet of a YAML pipeline configuration that includes both the Maven build step and a step to publish test results:

```
trigger:
  - main
pool:
  vmImage: 'ubuntu-latest'

steps:

  # Step 1: Maven Build and Test Execution

  - task: Maven@3

    inputs:

      mavenPomFile: 'pom.xml'  # Ensure your pom.xml is at the repository root

      goals: 'clean package'

      options: "            # Add any additional Maven options if needed
```

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```
# Step 2: Publish Unit Test Results
- task: PublishTestResults@2
  inputs:
    testResultsFiles: '**/target/surefire-reports/TEST-*.xml'
    mergeTestResults: true
    testRunTitle: 'Maven Unit Test Results'
```

Explanation of the YAML Steps:

1. **Maven Build Step:**
   - **Task:** Maven@3
   - **Input:**
     - **mavenPomFile:** Points to your pom.xml.
     - **goals:** The goals clean package ensure that your code is cleaned, compiled, tested, and packaged. The Surefire plugin runs tests during this process.
   - **Outcome:** Maven executes the tests and creates XML reports under the target/surefire-reports/ folder.
2. **Publish Test Results Step:**
   - **Task:** PublishTestResults@2
   - **Input:**
     - **testResultsFiles:** Uses a glob pattern (**/target/surefire-reports/TEST-*.xml) to find all XML test result files.
     - **mergeTestResults:** Set to true so that if there are multiple result files, they are merged into a single report.
     - **testRunTitle:** Provides a title for the test run that will appear in the Azure DevOps test results view.
   - **Outcome:** Azure Pipelines reads the test result XML files and displays a summary in the pipeline's **Tests** tab. You'll be able to see metrics like the number of tests passed, failed, or skipped, and review detailed error messages for any failures.

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3. Running and Verifying the Pipeline

After committing the YAML file to your repository, the pipeline is triggered (either automatically or manually):

1. **Trigger the Pipeline:**
   - Once your YAML file is saved in your repository, Azure Pipelines will pick up the changes. If not automatically triggered, you can click **"Run pipeline"** manually.
   - **Screenshot Tip:** Capture the pipeline run screen where you see the stages (e.g., Checkout, Maven Build, Publish Test Results).
2. **Monitor the Build Output:**
   - Navigate to the **"Build"** or **"Logs"** section of the pipeline run.
   - Confirm that the Maven task logs show the execution of tests and that the Surefire reports are created.
3. **Review Test Reports:**
   - Once the build completes, click on the **"Tests"** tab (often found on the pipeline summary page) to review the detailed test results.
   - **Screenshot Tip:** Capture the test report summary, showing the number of tests executed, passed, and failed. You may also capture detailed logs for any failed tests.

4. Troubleshooting Tips

- **No Test Results Found:**
  If you see a warning that no test results were found, verify that:
  - Your tests are indeed being executed (check the Maven build log).
  - The path in testResultsFiles correctly matches the location and naming pattern of the generated XML files.
- **Test Failures:**
  If tests fail, review the published test report for detailed error messages and stack traces. Adjust your tests or code as necessary and re-run the pipeline.

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Experiment 11: Creating Release Pipelines: Deploying Applications to Azure App Services, Managing Secrets and Configuration with Azure Key Vault, Hands-On: Continuous Deployment with Azure Pipelines

1. Overview

In this experiment, you will:

- **Deploy your build artifact** (e.g., a JAR or WAR file from a Maven/Gradle project) to an Azure App Service.
- **Manage secrets and configuration** securely using Azure Key Vault.
- **Set up a release pipeline** in Azure DevOps that automatically deploys your application when a new artifact is available (continuous deployment).

This experiment bridges the gap between build automation (CI) and release automation (CD) while ensuring secure management of sensitive information.

2. Prerequisites

Before you begin, ensure that you have:

- An **Azure DevOps account** with a project set up (see Experiment 9 and Experiment 10).
- A **build artifact** (e.g., your Maven/Gradle artifact) available from a build pipeline.
- An **Azure Subscription** with an **Azure App Service** instance already created to host your application.
- An **Azure Key Vault** instance created in your Azure Subscription for storing secrets (e.g., connection strings, API keys).
- Appropriate permissions to create and manage resources in Azure DevOps and your Azure Subscription.

3. Creating a Release Pipeline in Azure DevOps

A. Create a New Release Pipeline

1. **Log in to Azure DevOps:**

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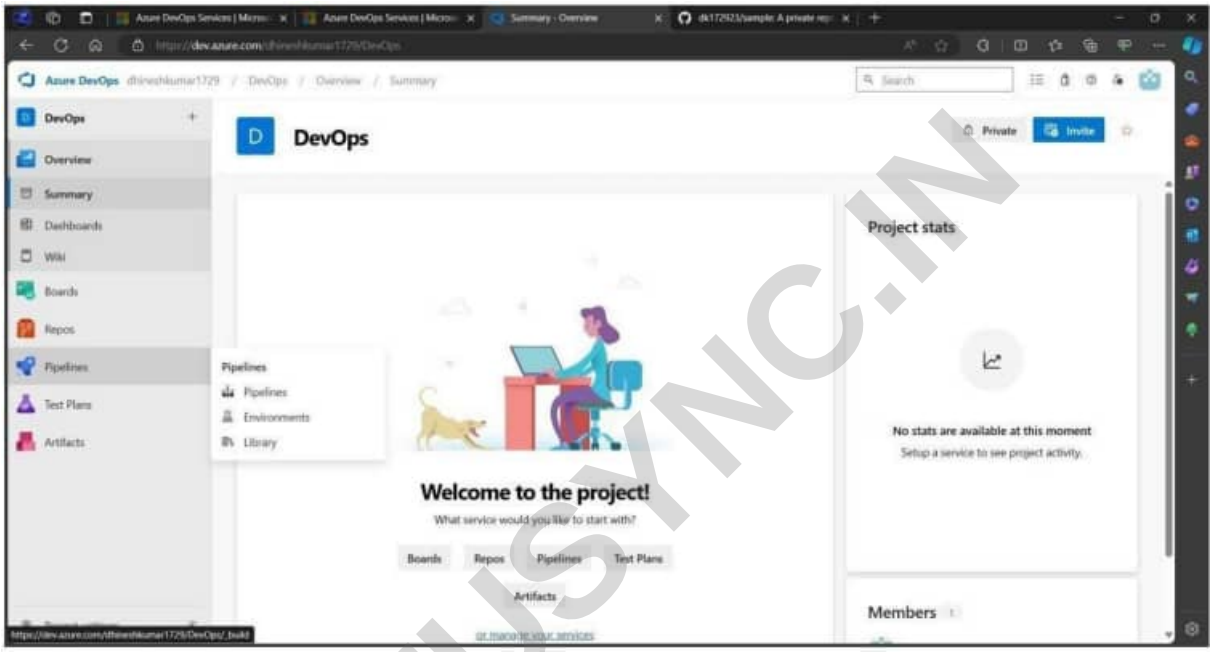
- Open your web browser and navigate to your Azure DevOps project (e.g., https://dev.azure.com/YourOrganization).

2. **Navigate to the Releases Section:**
   - In the left-hand menu, click on **"Pipelines"** and then **"Releases"**.
3. **Create a New Pipeline:**
   - Click on **"New pipeline"**.
   - When prompted, select **"Empty job"** (or start with a template if one suits your needs).



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B. Add an Artifact

1. **Link Your Build Artifact:**
   - In the release pipeline editor, click on **"Add an artifact"**.
   - **Source Type:** Choose the source (e.g., "Build").
   - **Source (Build Pipeline):** Select the build pipeline that produces your Maven/Gradle artifact.
   - **Default Version:** Use the latest version (or specify a branch/tag as needed).
   - Click **"Add"**.

C. Define a Stage for Deployment

1. **Add a New Stage:**
   - Click on **"Add a stage"** and select **"Empty job"**.
   - Rename the stage (e.g., Development or Production).
2. **Configure the Stage:**
   - Click on the stage to open its settings.
   - In the **"Tasks"** view for the stage, click **"+"** to add a new task.

4. Deploying to Azure App Services

A. Add the Azure App Service Deploy Task

1. **Add Deployment Task:**
   - In the stage tasks, click on **"+ Add"** and search for **"Azure App Service Deploy"**.
   - Click **"Add"** to include it in your stage.
2. **Configure the Azure App Service Deploy Task:**
   - **Azure Subscription:**
     - Click on **"Authorize"** (if required) and select your Azure subscription.
   - **App Service Type:** Choose **"Web App on Windows"** or **"Web App on Linux"** based on your deployment target.
   - **App Service Name:** Enter the name of your Azure App Service instance.
   - **Package or Folder:**
     - Specify the path to your build artifact (for example, $(System.DefaultWorkingDirectory)/_HelloMaven-CI/drop/HelloMaven-1.0-SNAPSHOT.jar or the folder containing your package). Adjust the path based on your artifact's location.
   - **Deployment Options:** Configure any additional options (e.g., deployment slot, resource group) as needed.

5. Managing Secrets and Configuration with Azure Key Vault

A. Create and Configure an Azure Key Vault (in Azure Portal)

1. **Create a Key Vault (if not already created):**
   - Log in to the Azure Portal.
   - Click on **"Create a resource"** and search for **"Key Vault"**.

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- Follow the prompts to create a new Key Vault (enter a name, select subscription, resource group, and region).

2. **Add Secrets to Your Key Vault:**
   - Once the Key Vault is created, navigate to it.
   - Click on **"Secrets"** and then **"Generate/Import"**.
   - Create new secrets (e.g., DBConnectionString, APIKey) and note their names.

B. Integrate Key Vault with Azure DevOps

1. **Create a Variable Group Linked to Key Vault:**
   - In your Azure DevOps project, navigate to **"Pipelines"** and then **"Library"**.
   - Click on **"+ Variable group"**.
   - Name the variable group (e.g., KeyVault-Secrets).
   - Enable **"Link secrets from an Azure Key Vault as variables"**.
   - Click **"Authorize"** and select your Azure subscription.
   - Select your Key Vault from the dropdown list.
   - Choose the secrets you want to import into the variable group.
   - Save the variable group.
2. **Use the Variable Group in Your Release Pipeline:**
   - In your release pipeline, click on the **"Variables"** tab.
   - Click **"Variable groups"** and then **"Link variable group"**.
   - Select the variable group you created (KeyVault-Secrets).
   - Now, you can reference these secrets as variables in your deployment tasks (for example, $(DBConnectionString)).

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6. Enabling Continuous Deployment

A. Configure Continuous Deployment Trigger

1. **Enable Continuous Deployment:**
   - In your release pipeline, click on the **"Triggers"** tab.
   - Under **"Artifact filters"** or **"Continuous deployment trigger"**, enable the toggle to allow the release pipeline to trigger automatically when a new build artifact is available.
   - Optionally, you can specify branch filters or artifact version filters.

2. **Save the Release Pipeline:**
    - Click **"Save"** to apply all changes.

B. Test the Continuous Deployment Pipeline

1. **Trigger a Build:**
    - Commit a change to your code repository to trigger your build pipeline (or trigger it manually).
    - Verify that the build pipeline creates a new artifact.
2. **Automatic Release:**
    - Once the new artifact is published, the release pipeline should automatically trigger a new release.
    - Monitor the release pipeline execution to ensure that:
        - The artifact is deployed to the Azure App Service.
        - The deployment task uses the configuration and secrets from Key Vault.

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Experiment 12: Practical Exercise and Wrap-Up: Build and Deploy a Complete DevOps Pipeline, Discussion on Best Practices and Q&A

1.Objective

To set up a CI/CD pipeline in Azure DevOps that continuously deploys an application to Azure App Services while securely managing configuration using Azure Key Vault.

2. Pre-requisites

- Azure Subscription
- Azure DevOps Account
- Azure CLI or Portal Access
- A sample application (e.g., Node.js, .NET)

3. Step-by-Step Instructions

A. Create Azure App Service

Go to Azure Portal (https://portal.azure.com)

Click Create a Resource → App Service

Choose: Resource Group, App Service Name, Runtime Stack (e.g., .NET, Node.js), Region

Click Review + Create

B. Create Azure Key Vault

Search for 'Key Vaults' in Azure Portal

Click + Create

Set the vault name and resource group

Click Review + Create

C. Store Secrets in Azure Key Vault

Go to your Key Vault → Secrets

Click + Generate/Import

Add secrets like: DbPassword, ApiKey

Set activation date and expiration (optional)

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D. Setup Azure DevOps Project & Repo

Go to Azure DevOps (https://dev.azure.com/)

Create a new project

Push your application code to the project's Repos

E. Create Azure Pipelines for CI/CD

Go to Pipelines → Create Pipeline

Choose your repo source

Use the classic editor or YAML

Add tasks: Build, Publish Artifact, Deploy to Azure App Service

F. Link Azure Key Vault to Pipeline

Go to Library in Azure DevOps Pipelines

Create a Variable Group

Link secrets from Azure Key Vault

Use these secrets in your pipeline using $(secretName)

G. Deploy to Azure App Service

Use the task Azure App Service Deploy

Select your Azure subscription and App Service

Set the deployment package path

Save and run the pipeline

4. Verification and Troubleshooting

- Verify deployment status in Azure DevOps Pipeline Logs
- Check Azure App Service URL for live application
- Inspect logs in App Service for runtime issues
- Ensure Key Vault access policies are correctly configured

5. Conclusion

You have successfully created an automated CI/CD release pipeline using Azure DevOps, securely managing application secrets through Azure Key Vault, and deploying to Azure App Services.

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